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**Post-MDT Oncology Workflow
Bottlenecks:
A Design-Science Study of an NHS
Integrated Data Repository and
Decision-Support Dashboard to
Reduce Delays from MDT Outcome to
Treatment Start.**

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Abstract

Delays between multidisciplinary team (MDT) decision-making and treatment initiation remain a significant challenge within National Health Service (NHS) cancer pathways. Operational data relevant to these workflows are frequently distributed across multiple systems, limiting visibility of bottlenecks and reducing the ability of service managers to intervene proactively. This study applied a Design Science Research (DSR) methodology to design, develop, and evaluate an Integrated Data Repository (IDR) and decision-support dashboard suite for post-MDT oncology pathway management within Portsmouth Hospitals University NHS Trust.

The artefact integrated oncology referral, outpatient clinic, and radiotherapy datasets using an automated extract-transform-load (ETL) architecture implemented in Python and PostgreSQL. A layered repository design incorporating staging, transformation, and analytics-ready reporting layers was developed to support pathway analysis and operational reporting. Three Power BI dashboards were produced to support referral management, radiotherapy service management, and end-to-end pathway monitoring.

Analysis of integrated pathway data identified a median post-MDT pathway duration of 37 days. The largest delay occurred between CT simulation and treatment initiation (16 days), followed by the interval between triage and first oncology clinic attendance (8 days). Repository development also identified previously unrecognised operational data-quality issues including duplicate referral records, incomplete clinic outcome recording, and inconsistencies between operational systems. The developed dashboards provided visibility of active pathway risks, radiotherapy capacity utilisation, and operational bottlenecks not readily observable through existing reporting processes.

The study contributes both a practical artefact and a set of evaluated design principles for operational oncology analytics. Findings demonstrate that meaningful improvements in operational visibility can be achieved through lightweight data integration approaches without replacing existing clinical systems. Future work should focus on integration of structured digital referrals, Systemic Anti-Cancer Therapy (SACT) datasets, clinic-capacity modelling, and further decomposition of treatment-planning workflows.

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Contents

Acknowledgements	2
List of Acronyms	8
1 Introduction	10
1.1 Problem Statement	11
1.2 Research Aim and Objectives	11
1.3 Research Approach	12
1.4 Academic and Practical Significance	13
2 Literature Review	14
2.1 Introduction	14
2.2 The NHS Cancer Pathway and MDT Framework	15
2.3 Operational Delays in Cancer Pathways	19
2.4 Data Fragmentation and the Need for Integrated Data Repositories	21
2.5 Dashboards, Operational Analytics, and Decision-Support Systems	23
2.6 Usability, HCI, and ISO 9241-110	25
2.7 AI-Enabled MDT Systems and Emerging Technologies	25
2.8 Synthesis of Gaps	26
2.9 Summary	28
3 Methodology	29

3.1	Overview of Research Design	29
3.2	Design Science Research Framework	31
3.3	Design and Development	32
3.4	Design Principles	40
3.5	Evaluation Strategy	42
3.6	Ethical and Governance Considerations	42
3.7	Summary	43
4	Results and Analysis	44
4.1	Introduction	44
4.2	Repository Development Results	44
4.3	Oncology Referral Analysis	45
4.4	Radiotherapy Service Analysis	47
4.5	End-to-End Pathway Analysis	49
4.6	Operational Evaluation	51
4.7	Evaluation of Design Principles	52
4.8	Summary	53
5	Discussion and Conclusions	55
5.1	Introduction	55
5.2	Discussion of Research Question 1	55
5.3	Discussion of Research Question 2	56
5.4	Artefact Evaluation	58
5.5	Evaluation of Design Principles	59
5.6	Reflections and Lessons Learned	60
5.7	Limitations	60
5.8	Future Work	61
5.9	Conclusion	63

References	64
A Integrated Data Repository Architecture	69
A.1 Repository Layers	69
A.2 Primary Repository Components	69
A.3 Source Systems	70
A.4 Logical Architecture	70
B Data Dictionary	71
C Example ETL and Business Rules	72
C.1 Identifier Standardisation	72
C.2 Referral Episode Consolidation	72
C.3 Incomplete Clinic Outcome Identification	73
C.4 Radiotherapy Pathway Status Classification	73
C.5 Canonical Pathway Events	73
D Dashboard Evaluation Results	74
D.1 Participants	74
D.2 System Usability Scale Results	74
D.3 Operational Value Assessment	74
D.4 Design Principle Validation	75
D.5 Qualitative Feedback	76

List of Figures

2.1	Iterative literate review process (Dawson, 2015)	16
2.2	Conceptual model linking literature-derived gaps, research objectives, design principles, artefact development, and evaluation	27
3.1	Logical Architecture of the IDR	35
3.2	Radiotherapy Pathway, Source Systems and Performance Milestones	38
4.1	Operational Oncology Referral Dashboard	45
4.2	Radiotherapy Capacity and Demand Dashboard	47
4.3	Departmental average number of fractions per linac (RAD20) for financial year 24/25 (NHS Digital, 2026b)	49
4.4	Integrated Post-MDT Pathway Dashboard	50

List of Tables

1.1	This mapping ensures methodological coherence between the research questions, artefact, and evaluation.	12
3.1	Source Systems Summary	33
3.2	Artefact to Research Objective mapping	40
4.1	Median duration between canonical events in the Radiotherapy post-MDT pathway	49
4.2	Artefact evidence of Design Principles	53
A.1	Repository Architecture Components	69
A.2	Primary Data Sources	70
B.1	Data Dictionary	71
D.1	Evaluation Participants	74
D.2	System Usability Scale (SUS) Results	74
D.3	Operational Value Assessment	75
D.4	Design Principle Evaluation Results	75
D.5	Summary of Qualitative Feedback	76

List of Acronyms

AI	Artificial Intelligence
CDSS	Clinical Decision Support System
CDW	Clinical Data Warehouse
CWT	Cancer Wait Time
DSR	Design Science Research
DSRM	Design Science Research Methodology
DTI	Diagnosis to Treatment Interval
DTT	Decision-to-Treat
ETL	Extract-Transform-Load
EPR	Electronic Patient Record
FDS	Faster Diagnosis Standard
FHIR	Fast Healthcare Interoperability Resources
HCI	Human Computer Interaction
IDR	Integrated Data Repository
JCCO	Joint Council for Clinical Oncology
MDT	Multidisciplinary Team
NCWTMDS	National Cancer Waiting Times Monitoring Data Set
NHS	National Health Service
OIS	Oncology Information System
PAS	Patient Administration System
PTL	Patient Tracking List
QA	Quality Assurance
RCR	Royal College of Radiologists
RTT	Referral To Treatment
SABR	Stereotactic Ablative Body Radiotherapy
SQL	Structured Query Language
SUS	System Usability Scale

Chapter 1

Introduction

Delivering timely cancer care remains a critical priority within the NHS, where delays at any stage of the diagnostic and treatment pathway can negatively affect patient outcomes, experience, and service performance. Recent policy reforms have consolidated the NHS Cancer Waiting Time (CWT) framework into three outcome-focused standards: the 28-day Faster Diagnosis Standard (FDS), the 31-day decision-to-treat to treatment (DTT) standard, and the 62-day referral-to-treatment (RTT) standard—implemented nationally from October 2023 (Johnson, 2024; NHS Digital, 2026a). These standards place increased emphasis on operational viability, rapid decision-making, and efficient progression of patients through the post-MDT (multidisciplinary team) stages of the oncology pathway. However, many Trusts continue to face systemic challenges in generating timely, accurate, and actionable insights from fragmented data sources.

Within Portsmouth Hospitals University NHS Trust, as in many NHS institutions, key operational data concerning MDT outcomes, clinic scheduling, booking practices, and treatment initiation remain dispersed across siloed systems. Although these systems individually support MDT coordination, outpatient clinic management, and treatment delivery, they lack a consolidated view of pathway performance. This fragmentation makes it difficult for operational managers to detect bottlenecks, identify emerging breach risks, or allocate clinical capacity efficiently. As a result, delays between **MDT outcomes**, **first oncology clinic**, **decision-to-treat**, and **first definitive treatment** remain insufficiently visible, impairing the Trust’s ability to monitor compliance with the 31- and 62-day standards and to respond rapidly to operational pressures.

1.1 Problem Statement

The central problem addressed in this research is the absence of an integrated, operationally focused data infrastructure and decision-support interface to illuminate post-MDT workflow delays and guide timely interventions within an NHS oncology service. Despite the availability of relevant data, the lack of integration and harmonisation across MDT, clinic, and treatment systems results in manual workarounds, inconsistent performance monitoring, and delayed insights. This gap diminishes the ability of operational teams to identify patients at risk of breaching national cancer standards, understand the factors driving variability in pathway intervals, and proactively manage capacity-constrained environments.

Moreover, existing reporting solutions tend to focus on aggregate compliance metrics after the fact, rather than providing real-time or near-real-time visibility of operational bottlenecks. Without a unified source of truth that aligns event definitions, timestamps, and clinical coding standards (NHS Digital, 2026a), it becomes difficult to execute performance analysis or support day-to-day decision-making. Literature on Integrated Data Repositories (IDRs) similarly highlights that many institutions struggle with data harmonisation, terminological alignment, and the creation of analytics-ready architectures capable of supporting both research and operational use cases (Gagalova *et al.*, 2020).

This study therefore seeks to design, develop, and evaluate a **lightweight Integrated Data Repository (IDR)** and accompanying **decision-support dashboard** tailored specifically to the post-MDT oncology workflow. By integrating key pathway events, clinic capacity information, and national standard definitions, the artefact aims to surface bottlenecks, highlight breach risks, and support operational teams in prioritising actions that improve pathway performance.

1.2 Research Aim and Objectives

1.2.1 Research Aim

To design and evaluate an IDR and decision-support dashboard capable of improving operational visibility and proactive management of post-MDT oncology pathways.

1.2.2 Objectives

1. Identify and quantify post-MDT pathway delays.
2. Determine operational factors associated with delay.
3. Design an integrated data architecture.

4. Develop a user-centred dashboard.
5. Evaluate utility, usability and technical validity.
6. Derive transferable design principles

Research Question	DSR Activity	Primary Methods
RQ1: Where do delays occur between MDT outcome and treatment?	Problem identification	Descriptive statistics, interval analysis
RQ2: What operational factors are associated with delay?	Problem analysis	Descriptive and operational analysis
RQ3: Does the artefact improve operational decision-making?	Evaluation	Task-based testing, time-to-answer, accuracy
RQ4: What design principles support post-MDT operational analytics?	Design knowledge	Literature synthesis and evaluation synthesis

Table 1.1: This mapping ensures methodological coherence between the research questions, artefact, and evaluation.

1.3 Research Approach

This study adopts Design Science Research (DSR) because the primary objective is not solely to explain an existing phenomenon but to create and evaluate an artefact intended to address an identified organisational problem. Whereas case-study research focuses on understanding contextual factors and Action Research emphasises organisational change through iterative intervention, DSR provides a structured framework for the design, construction and evaluation of innovative technological artefacts while simultaneously generating transferable knowledge (Hevner *et al.*, 2004; Peffers *et al.*, 2007).

The approach is particularly appropriate given the need to integrate heterogeneous operational data sources and translate them into actionable decision support for NHS oncology services. Similar DSR approaches have been successfully applied within health informatics, clinical decision support, and healthcare analytics domains where practical solutions and theoretical contribution are required concurrently (Harper and Mustafee, 2023; Johnson *et al.*, 2020).

The study therefore follows the six-stage Design Science Research Methodology (DSRM) proposed by Peffers *et al.* (2007), with detailed application described in Chapter 3.

1.4 Academic and Practical Significance

Academically, the research contributes to the limited body of DSR-grounded studies focusing on NHS operational analytics, offering:

- an IDR reference architecture for oncology operations;
- a set of design principles for NHS-appropriate decision-support dashboards;
- an evaluation protocol combining usability metrics, decision-quality testing, and technical verification.

Practically, the artefact provides operational teams with a tool that can:

- reduce time-to-insight;
- support rapid identification of bottlenecks and breach risks;
- align capacity planning with demand;
- improve pathway monitoring in line with the 28-, 31-, and 62-day standards.

Ultimately, the study aims to support faster, more reliable progression from MDT decision to treatment initiation, with the potential to support improved patient care and more effective management of oncology services.

Chapter 2

Literature Review

2.1 Introduction

The purpose of this literature review is to synthesise and critically evaluate existing research relevant to the design of an Integrated Data Repository (IDR) and operational analytics dashboard aimed at improving post-MDT oncology pathway efficiency. Timely cancer treatment has profound implications for survival, patient experience, resource utilisation, and system performance. Evidence from national datasets, observational studies, systematic reviews, and informatics research shows that delays between MDT decisions and treatment initiation remain persistent across the NHS. Furthermore, data fragmentation, inconsistent recording practices, and limited visibility of operational bottlenecks hinder the ability of clinicians, MDT coordinators, and service managers to proactively manage these delays.

The review is organised around five interrelated themes:

1. MDT operations and evolving pressures on multidisciplinary cancer decision-making;
2. Operational delays across the cancer pathway, particularly diagnosis-to-treatment intervals (DTI);
3. Data fragmentation and the case for IDR's;
4. Dashboards and performance-monitoring tools to support operational decision-making;
5. Emerging data trends and the role of ETL, interoperability, AI and multimodal data integration.

Collectively, this literature forms the conceptual foundation (kernel theory) for the artefact developed in this study. The following sections provide a detailed account of these themes, integrating high-quality evidence from the UK, international datasets, clinical oncology, health informatics, and organisational science.

2.1.1 Method

An iterative literature review was adopted, following the guidance outlined by Dawson (2015) (Figure 2.1). An initial search was conducted using Google Scholar and open-access repositories (arXiv.org) to identify foundational concepts and terminology related to IDR’s, cancer pathways, MDT operations, dashboards, and Design Science Research. These preliminary results informed a more structured search strategy applied to ScienceDirect, IEEEExplore, and PubMed.

Searches were limited to peer-reviewed journals, systematic reviews, national audits, and authoritative policy documents published primarily between 2010 and 2026, with earlier seminal works retained where conceptually relevant (e.g. JCCO report (Ash *et al.*, 2004), the Calman-Hine report (Calman and Hine, 1995) and Design Science Research foundations). Studies were included if they addressed MDT processes, oncology pathway delays, health data integration, dashboards, usability standards, or operational analytics. Articles focused exclusively in molecular biology, single-modality clinical trials, or non-healthcare data systems were excluded.

Approximately 90 sources were reviewed during the process, with around 55 core studies retained for detailed analysis based on relevance, methodological quality, and applicability to NHS operational contexts. Backward citation tracking was used to identify influential seed papers and ensure conceptual completeness.

2.2 The NHS Cancer Pathway and MDT Framework

2.2.1 MDTs as the Backbone of UK Cancer Care

Since the publication of the Calman–Hine report (Calman and Hine, 1995) and subsequent interventions, MDTs have been embedded as the central governance structure for cancer decision-making (Haward, 2006; Price *et al.*, 2014; Morris *et al.*, 2006, 2008). The MDT model mandates that every new cancer case in the UK is discussed by a multidisciplinary panel including oncologists, surgeons, radiologists, pathologists, specialist nurses, and allied professionals. The 2024 NHS Wales MDT Charter emphasises the crucial role MDTs play in standardising care, reducing unwarranted variation, and improving outcomes by bringing domain experts together in real time (NHS Wales, 2024). Prospective benefits include higher adherence to guidelines, improved communication, shared decision-making, and better coordination of multi-modal treatment plans.

However, the complexity of modern cancer care has increased dramatically alongside rising incidence, new treatment options, biomarker-based stratification, digital imaging volume, and genomic data availability. MDTs now contend with substantially larger case numbers, more

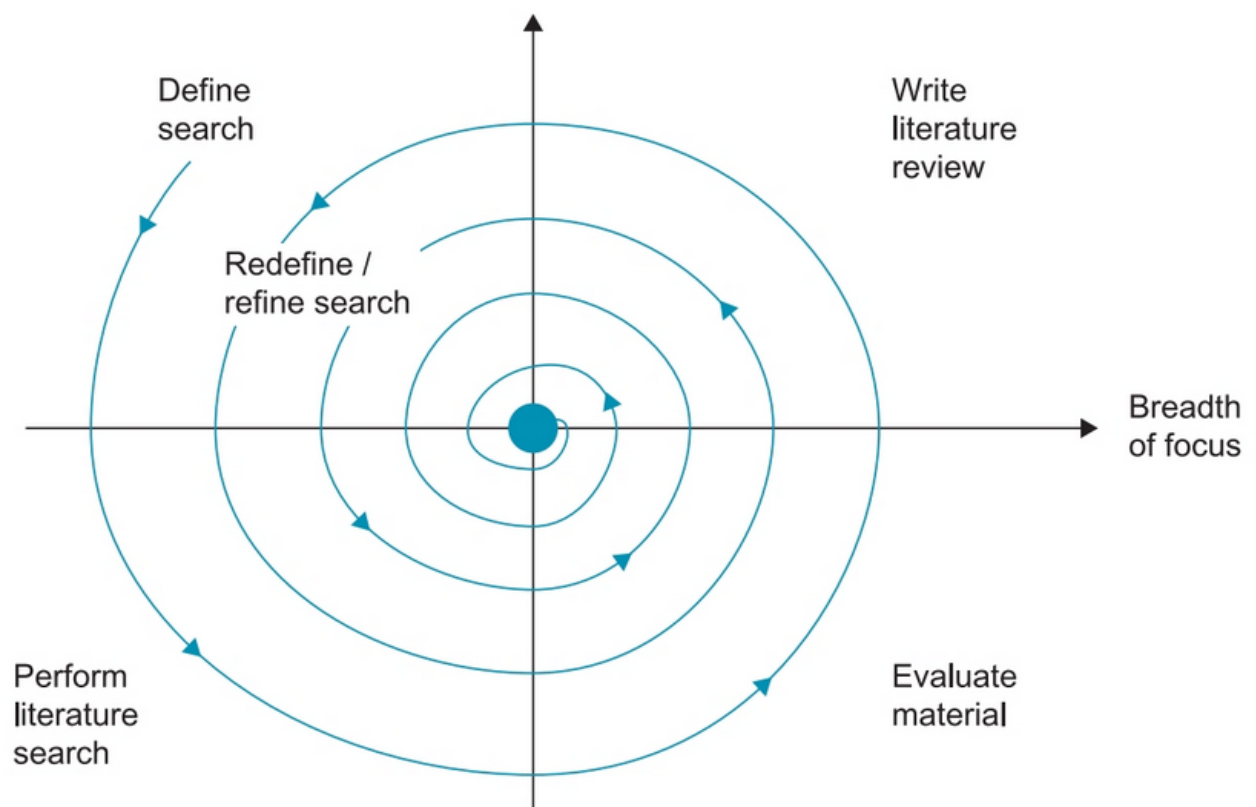


Figure 2.1: Iterative literature review process (Dawson, 2015)

complex diagnostic information, and resource pressures that stretch meeting capacity. Professional societies and national audits consistently highlight that MDT meetings are often over-subscribed, under-resourced, and administratively burdensome. These pressures compromise the quality of discussion and create operational bottlenecks that directly affect patient pathways.

2.2.2 Evidence of MDT Operational Pressures

Collectively, the literature demonstrates that MDTs are experiencing growing operational pressure resulting from increasing cancer incidence, expanding diagnostic complexity, and constrained service capacity. Across multiple studies, common themes emerge including administrative inefficiencies, information deficits, scheduling pressures, and resource limitations.

Soukup *et al.* (2022) identified logistical challenges affecting almost one-third of MDT discussions, while Law *et al.* (2024) highlighted persistent barriers relating to communication, attendance, information availability, and workload. These findings suggest that operational inefficiencies are not isolated incidents but recurring characteristics of contemporary MDT environments.

Evidence from pathway-performance studies reinforces this position. Lim *et al.* (2025) reported poor national compliance with cancer waiting-time standards across UK head and neck services, while Chakravarty *et al.* (2023) demonstrated that additional complexity within referral pathways was associated with significantly longer treatment delays. Together, these studies indicate that operational challenges extend beyond MDT meetings themselves and influence downstream stages of the cancer pathway.

A recurring theme across the literature is therefore the lack of visibility between MDT decision-making and subsequent treatment delivery. While studies identify operational pressures and poor pathway performance, relatively little attention has been given to how integrated operational data might support earlier identification of emerging bottlenecks.

While these studies consistently demonstrate operational strain within MDT processes, several limitations should be acknowledged. Much of the existing evidence is observational and focuses primarily on meeting effectiveness, communication quality, or decision-making processes rather than downstream operational consequences of MDT outcomes. For example, Soukup *et al.* (2022) identify logistical and administrative barriers that impair MDT performance, yet the study does not examine whether these issues translate into measurable delays in treatment initiation. Similarly, the scoping review proposed by Law *et al.* (2024) aggregates a broad range of organisational challenges but provides limited insight into which barriers exert the greatest influence on pathway performance.

Large-scale cohort studies such as Lim *et al.* (2025) quantify poor compliance with national

standards, but their retrospective design offers little visibility of the specific workflow stages responsible for delay accumulation. Consequently, while the literature establishes that MDT processes operate under significant pressure, a gap remains in understanding how operational data can be used to identify, monitor, and mitigate delays arising after MDT decisions have been made. This distinction is important because improving MDT discussion quality alone may be insufficient if subsequent scheduling and treatment processes remain poorly coordinated.

2.2.3 National policy direction and waiting time standards

NHS England’s 2024/25 planning guidance emphasises compliance with the Faster Diagnosis Standard (FDS) and cancer treatment timeliness metrics. Persistent waiting-time challenges, data-quality requirements, and the need for integrated pathway oversight tools are highlighted (Parker, 2026). Radiotherapy services face particular strain: the Royal College of Radiologists reported over 2,300 breaches of the 31-day standard in October 2023 (RCR, 2024). Hidden waits—such as 20-week delays between surgery and radiotherapy—are not systematically captured, further illustrating limitations in operational visibility.

2.2.4 Radiotherapy Waiting-Time Recommendations

In addition to national Cancer Waiting Time standards, radiotherapy services have historically been supported by recommendations published by the Joint Council for Clinical Oncology (JCCO). The JCCO framework recognises that clinically meaningful delay may occur within treatment planning and booking processes that are not always visible within national waiting-time metrics. Ash *et al.* (2004) demonstrated substantial variation in radiotherapy waiting times across the United Kingdom and highlighted the importance of monitoring elapsed time between referral and treatment delivery.

Unlike national cancer waiting-time standards, which primarily support performance monitoring and accountability, JCCO recommendations provide a clinically focused perspective on the timeliness of radiotherapy delivery. This distinction is important because patients may remain compliant with national targets while still experiencing delays that affect operational performance and service quality. Consequently, both national standards and clinically derived radiotherapy targets have relevance when designing operational analytics for radiotherapy services.

2.3 Operational Delays in Cancer Pathways

2.3.1 Diagnosis-to-Treatment Interval (DTI) as a Critical Metric

A robust body of literature demonstrates that prolonged DTI adversely affects survival across multiple cancer types.

Key findings include:

- Hanna *et al.* (2020): A systematic review of 34 studies (1.27M patients) found each four-week delay increased mortality by up to 6–8% across treatment modalities.
- Coca-Pelaz *et al.* (2018): Delays in Head and Neck Small Cell Carcinoma beyond four weeks contribute to tumour progression and reduced local control.
- Sharma *et al.* (2016): Delays greater than 30 days reduce overall survival (HR 1.12), with mortality increasing 2.2% per additional week.
- Villemure-Poliquin *et al.* (2025): Treatment within 30 days improves survival by 9%.
- Saleh *et al.* (2024): Cervical cancer survival drops dramatically after 30 days.

Despite differences in cancer type, geography, and study design, the evidence consistently demonstrates that treatment delay is associated with poorer clinical outcomes. Systematic reviews and large cohort studies identify remarkably similar trends, with survival deteriorating as diagnosis-to-treatment intervals increase. However, the magnitude of effect varies between tumour groups, suggesting that pathway urgency may differ according to tumour biology and disease progression rates. This consistency across studies strengthens the argument that reducing operational delay is clinically important, while variation in effect size highlights the need for pathway-specific operational management strategies.

2.3.2 Hospital/System Factors Influencing Delay

Across the reviewed studies, organisational influences on delay can be grouped into three broad categories: diagnostic bottlenecks, administrative coordination challenges, and capacity constraints. Diagnostic delays frequently arise from imaging and biopsy availability, while administrative delays result from scheduling inefficiencies, incomplete information, or pathway handovers. Capacity-related delays emerge where demand exceeds available clinic, theatre, chemotherapy, or radiotherapy resources. Although studies differ in emphasis, all point towards organisational processes rather than purely clinical factors as major contributors to pathway delay.

A further limitation of the current literature is the tendency to treat organisational performance as a static characteristic rather than a dynamic process. Factors such as hospital volume,

resource availability, and service configuration are frequently associated with treatment delays; however, there is comparatively little exploration of how these risks evolve over time or how operational teams can intervene before delays become breaches. This reinforces the distinction between retrospective performance measurement and proactive pathway management.

2.3.3 MDT-specific drivers of post-decision delays

- Missing diagnostic information and administrative errors cause re-discussion delays (Soukup *et al.*, 2022).
- High caseloads reduce decision completeness.
- Secondary-care referrals introduce additional diagnostic delays (Chakravarty *et al.*, 2023).
- Capacity constraints (radiotherapy, theatres, oncology clinics) generate downstream bottlenecks.

Although the evidence linking treatment delay and mortality is compelling, caution should be exercised when applying these findings directly to operational improvement initiatives. Most studies evaluate aggregated diagnosis-to-treatment intervals and therefore provide limited insight into the relative contribution of individual pathway stages. While the findings clearly establish delay as clinically important, they do not identify where operational interventions are likely to deliver the greatest benefit. From a service management perspective, this highlights the need for more granular operational analytics capable of decomposing pathway delays into actionable components.

2.3.4 Critical synthesis of delay literature

Collectively, the literature provides strong evidence that prolonged diagnosis-to-treatment intervals adversely affect patient outcomes across multiple cancer types. However, important limitations remain when attempting to translate this knowledge into operational practice. Most studies conceptualise delay as a single aggregated interval and consequently offer limited support for identifying specific workflow bottlenecks. Furthermore, while organisational variables such as hospital volume, diagnostic complexity, and service capacity are frequently identified as contributors to delay, relatively little attention has been given to the role of integrated operational data in supporting proactive intervention.

This omission is particularly significant within NHS oncology services, where national performance standards require action before breaches occur. Once a patient has experienced a clinically significant delay, retrospective performance reporting provides little opportunity to prevent harm. Accordingly, there is a need for analytical approaches capable of identifying emerging risks and supporting early operational intervention, rather than simply documenting historical performance.

2.4 Data Fragmentation and the Need for Integrated Data Repositories

2.4.1 Fragmentation Across NHS Systems

Oncology data spans EPRs, PAS, pathology, RIS/PACS, radiotherapy systems, chemotherapy e-prescribing, Somerset/Cosmic, and MDT spreadsheets.

Evidence includes:

- Operational policy from Royal Wolverhampton highlights inconsistent event definitions and PTL challenges (The Royal Wolverhampton NHS Trust, 2024).
- Varma *et al.* (2026): 92.3% agreement between CWT and hospital data only when treatment occurs within 100 days; discrepancies arise from definition mismatches and missing fields.

2.4.2 Clinical Data Warehouses (CDWs) and the Evolution Toward IDRs

Examples:

- ROOT CDW: real-time repository for 67,000+ cancer patients (Jung *et al.*, 2021).
- ONCO-FAIR: chemotherapy-specific FHIR extensions (Guilbert *et al.*, 2024).
- NHS–industry multimodal genomic-clinical data lake (Conway *et al.*, 2025).
- HN-XNAT: federated imaging and radiotherapy repository (Butterworth *et al.*, 2025).

Despite differing objectives, several common characteristics can be observed across these repository implementations. All seek to integrate heterogeneous oncology datasets, establish consistent data models, and improve data accessibility for downstream analysis. However, important differences also exist. ROOT prioritises real-time operational data aggregation, whereas ONCO-FAIR focuses on interoperability and standards compliance through FHIR extensions. The NHS–industry data lake emphasises large-scale multimodal research integration, while HN-XNAT adopts a federated model centred on imaging and radiotherapy data.

Collectively, these implementations demonstrate that successful repositories require data harmonisation, governance frameworks, and scalable integration architectures. However, few have been designed specifically to support operational cancer pathway management. This distinction directly informs the proposed artefact, which prioritises workflow visibility, breach-risk identification, and operational decision support rather than research-oriented data reuse.

While these examples demonstrate the technical feasibility of large-scale oncology data repos-

itories, their transferability to routine NHS operational environments warrants careful consideration. Many reported implementations originate from digitally mature organisations with substantial informatics capability, dedicated technical teams, and highly structured data environments. Such conditions may not be representative of NHS Trusts where operational data are frequently distributed across legacy systems, locally developed databases, spreadsheets, and departmental applications.

Furthermore, the majority of published repositories prioritise research, population health, or precision medicine objectives. Although these use cases require extensive data integration, they often place less emphasis on the timeliness, operational responsiveness, and workflow visibility required for day-to-day pathway management. Consequently, a gap exists between the design of many clinical data repositories and the practical needs of operational cancer services.

2.4.3 Need for IDRs in NHS Operational Oncology

The absence of a robust IDR has implications beyond simple reporting inefficiency. Fragmented datasets create multiple versions of the truth, requiring operational staff to reconcile conflicting information from different systems before decisions can be made. This process introduces delays, increases administrative burden, and reduces confidence in performance measures.

Moreover, fragmented data environments encourage reactive management practices. Operational teams frequently identify pathway risks only after breaches have occurred because no single system provides a comprehensive view of patient progression across MDT, clinic, and treatment stages. As a result, opportunities for early intervention may be missed despite the underlying data already existing within organisational systems.

2.5 Dashboards, Operational Analytics, and Decision-Support Systems

2.5.1 The Role of Dashboards in Performance Management

Dashboards synthesise high-volume data for actionable insight. NHS guidance emphasises dashboards as essential for meeting FDS and treatment targets.

Buttigieg *et al.* (2017) identify:

- Strategic dashboards,
- Tactical dashboards,
- Operational dashboards.

Key design principles include real-time refresh, drill-downs, user-centred design, and integration with data infrastructures.

Despite widespread adoption, dashboards are not universally associated with improved organisational performance. Several studies have reported that dashboards frequently become passive reporting tools rather than active decision-support systems. The mere presentation of information does not guarantee action, particularly when users are confronted with excessive metrics, poorly aligned visualisations, or data that lack clear operational relevance.

Consequently, dashboard effectiveness depends not only on technical implementation but also on the extent to which information is embedded within operational decision-making processes. This distinction is particularly relevant in cancer pathway management where users require actionable intelligence capable of informing scheduling, capacity allocation, and escalation decisions.

2.5.2 Clinical Decision-Support Dashboards and Multimodal Data

Examples:

- Multimodal CDSS supporting 170,000+ patients, using ETL + NLP + 140 data quality checks (Chang *et al.*, 2025).
- ROOT CDW: automated ingestion, structured/unstructured data integration (Jung *et al.*, 2021).

2.5.3 Comparative Analysis of Dashboard Implementations

The reviewed dashboard implementations share several common design objectives, including improved information visibility, support for timely decision-making, and reduction of administrative burden. Both ROOT and multimodal CDSS environments rely upon integrated data architectures to provide trustworthy analytics, emphasising the dependency between dashboard performance and underlying data quality.

However, notable differences exist regarding intended usage. Clinical decision-support dashboards are primarily oriented towards patient-level treatment decisions, whereas operational dashboards focus on service performance, capacity management, and workflow optimisation. The latter perspective receives comparatively less attention within the literature despite its importance for achieving national waiting-time standards.

These observations suggest that an effective oncology operational dashboard should combine characteristics of both approaches: robust data integration, user-centred visualisation, and proactive decision-support capabilities tailored to pathway management.

2.5.4 Data Quality and Informatics Challenges

Dashboards depend on strong data foundations:

- Coding variation limits dataset reliability (Varma *et al.*, 2026).
- Oncology workloads require custom FHIR extensions (Guilbert *et al.*, 2024).
- Multimodal data lakes require governance and standardisation (Conway *et al.*, 2025).

2.5.5 Reactive versus Proactive Operational Management

Traditional healthcare performance reporting has largely focused on retrospective measurement through key performance indicators and compliance monitoring. While such approaches provide valuable accountability mechanisms, they are inherently limited in their ability to prevent adverse outcomes because they identify problems only after they have occurred.

This limitation is particularly relevant in oncology services. Evidence demonstrates that delays in treatment initiation are associated with poorer patient outcomes (Hanna *et al.*, 2020; Sharma *et al.*, 2016). Consequently, a management approach based solely on retrospective review is problematic because a missed cancer target represents more than a performance failure; it may indicate a clinically significant delay experienced by a patient.

Continuous improvement frameworks such as Plan-Do-Check-Act (PDCA) remain widely used within healthcare quality improvement. However, PDCA assumes that outcomes can be observed, measured, and refined through successive iterations. In cancer pathway management, allowing delays to occur before evaluating performance may be ethically and operationally undesirable. Instead, there is a need for leading indicators capable of identifying emerging risks before breaches occur.

From an operational analytics perspective, this shifts the emphasis from historical reporting towards predictive and proactive decision support. Dashboards designed around breach-risk identification, remaining time-to-target, and capacity constraints may therefore provide greater practical value than systems focused exclusively on retrospective compliance measurement.

2.6 Usability, HCI, and ISO 9241-110

ISO 9241-110:2020 (ISO9241, 2020) provides seven principles:

1. Suitability for the task
2. Self-descriptiveness
3. Conformity with user expectations

4. Learnability
5. Controllability
6. Error tolerance
7. Suitability for individualisation

Fraunhofer usability research supports its relevance in healthcare (Hunkirchen, n.d.).

2.7 AI-Enabled MDT Systems and Emerging Technologies

Examples include:

- EvoMDT: multi-agent MDT recommender reducing decision time by 30–40% (Liu *et al.*, 2026a).
- TrustedMDT: TNM staging and guideline checking integrated with Teams (Soltan *et al.*, 2025).
- AI-MDT for lung cancer: improved throughput and workload reduction (Liu *et al.*, 2026b).
- AI automation of transcription and summarisation (HealthOrbit, 2025).
- Multi-agent consensus MDT system achieving high inter-agent agreement (Han *et al.*, 2025).

While AI-enabled MDT systems report promising improvements in efficiency, consistency, and decision support, many studies remain limited by controlled implementation environments and highly structured datasets. Questions therefore remain regarding external validity and scalability within routine NHS settings characterised by fragmented information systems and variable data quality.

Importantly, the effectiveness of AI applications appears dependent upon the existence of integrated, standardised, and high-quality data architectures. Consequently, AI may be better viewed as a downstream capability rather than a standalone solution. Without addressing foundational challenges relating to interoperability, governance, and data integration, many of the anticipated benefits of AI-assisted MDT decision-making may remain difficult to realise in practice.

2.8 Synthesis of Gaps

The literature reveals a common pattern across otherwise diverse research domains. Studies examining MDT performance identify increasing operational pressure but provide limited visibility of downstream workflow delays. Research linking treatment delays to survival demonstrates the importance of timely treatment but offers limited guidance regarding how delays should be managed operationally. Similarly, data repository research demonstrates integration capability while dashboard studies emphasise information presentation; however, few studies combine these strands into a coherent operational solution.

The literature review identifies four principle gaps.

Gap 1: Limited visibility of post-MDT workflow delays. While existing studies identify treatment delays and MDT inefficiencies, few provide mechanisms for analysing delay accumulation between MDT outcome and treatment initiation. This gap directly informs Objective 1 (identify and quantify post-MDT pathway delays) and Objective 2 (determine operational factors associated with delay).

Gap 2: Predominantly reactive performance management. Most existing reporting approaches focus on retrospective compliance measurement rather than proactive intervention. This gap informs DP1 (Surface Breach Risks as a Leading Indicator) and DP2 (Pair Risk Information with Actionable Capacity Data).

Gap 3: Limited operational application of Integrated Data Repositories. Existing repository research focuses primarily on research, genomics, and population health use cases rather than pathway operations. This gap directly supports Objective 3 (design an integrated data architecture) and DP3 (Use Canonical Standard-Aligned Pathway Events).

Gap 4: Insufficient evidence regarding user-centred operational dashboards. Although dashboard research demonstrates value for healthcare monitoring, limited attention has been given to oncology pathway management and task-oriented operational decision support. This gap informs Objective 4 (develop a user-centred dashboard) and DP6 (Align Interaction Design with Real Operational Tasks).

Collectively, these gaps establish the need for an artefact that combines integrated operational data, proactive pathway monitoring, and user-centred decision support. These requirements form the foundation of the Design Science Research approach outlined in Chapter 3.

Figure 2.2 synthesises the findings of the literature review and illustrates how the identified research gaps inform the objectives and design principles underpinning the artefact. The model demonstrates that challenges relating to MDT operational pressures, pathway delays, data fragmentation, reactive management practices, and usability deficiencies collectively motivate the development of an IDR and decision-support dashboard. These gaps are translated into

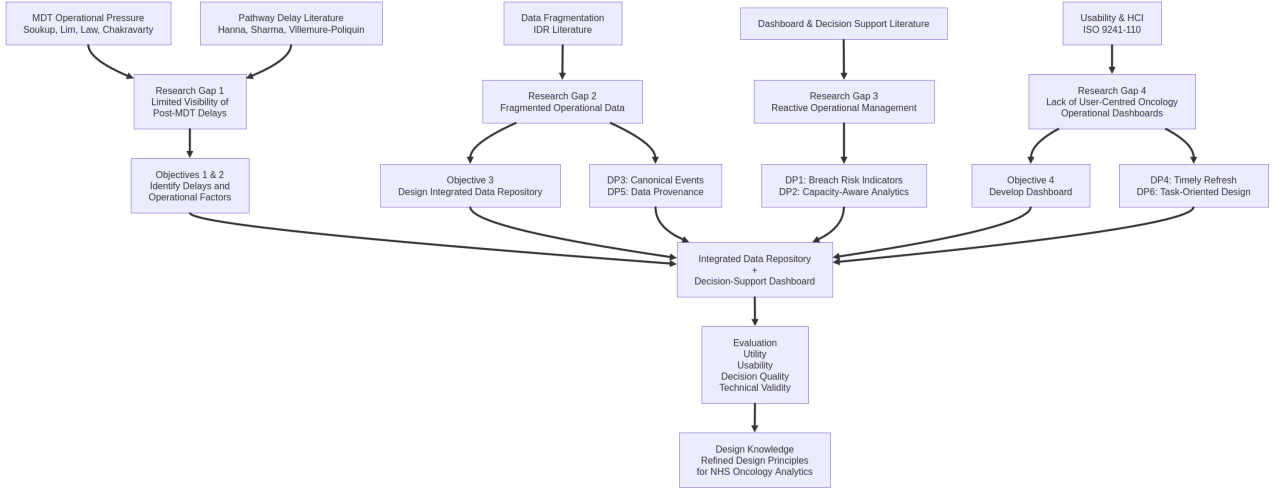


Figure 2.2: Conceptual model linking literature-derived gaps, research objectives, design principles, artefact development, and evaluation

explicit objectives and design principles which guide artefact development and subsequently form part of the evaluation framework. The resulting design knowledge contributes both a practical solution for NHS oncology services and transferable principles for operational analytics in healthcare.

2.9 Summary

This literature review synthesises evidence across MDT functioning, treatment delays, operational workflow challenges, data fragmentation, health informatics, dashboard design, usability standards, and emerging AI-driven MDT tools. MDTs remain vital but increasingly pressured. Delays between diagnosis, MDT decisions, and treatment initiation significantly worsen outcomes, yet Trusts lack the integrated systems required to manage these pathways proactively. Modern CDWs demonstrate feasible architectures, while dashboard and usability literature highlight design imperatives. AI-enabled MDT systems reveal future potential but highlight the foundational need for high-quality integrated data repositories.

The combined evidence exposes a clear research gap: NHS oncology services require a robust, user-centred, IDR-backed operational dashboard to improve MDT-to-treatment coordination.

Chapter 3

Methodology

3.1 Overview of Research Design

This research adopts a Design Science Research (DSR) methodology to address an operational problem in NHS oncology services: the lack of integrated, actionable visibility of delays between MDT outcomes and treatment initiation. DSR is particularly suitable for this study because it explicitly focuses on the design, construction, and evaluation of an artefact intended to improve a real-world organisational context, while generating transferable design knowledge (Hevner *et al.*, 2004).

The primary artefact developed in this study is a lightweight Integrated Data Repository (IDR) and decision-support dashboard, designed to consolidate post-MDT pathway data and support operational decision-making. In addition to the artefact itself, the research produces a set of evaluated design principles for oncology operational analytics, constituting a contribution to knowledge beyond the local implementation.

3.1.1 Research Philosophy and Methodological Positioning

The methodological choices adopted in this study can be understood using the Research Onion framework proposed by Saunders *et al.* (2023), which emphasises alignment between research philosophy, methodological approach, strategy, and methods.

At the philosophical level, this study adopts a pragmatic worldview. Pragmatism is concerned primarily with addressing real-world problems and selecting methods that are most appropriate for answering the research questions rather than adhering strictly to either positivist or interpretivist traditions. This orientation is particularly appropriate because the study seeks both to understand operational challenges within post-MDT oncology pathways and to develop a practical technological solution capable of addressing those challenges.

Pragmatism aligns naturally with Design Science Research (DSR), where knowledge is generated through the construction and evaluation of artefacts within a real organisational context. The focus of the present study is therefore not simply explanation of existing phenomena, but the creation of actionable knowledge through development and evaluation of an Integrated Data Repository and decision-support dashboard.

At the methodological level, the research adopts an abductive approach. Existing literature relating to oncology workflow delays, data integration, operational analytics, and dashboard design informed initial design requirements and design principles. These concepts were subsequently refined through iterative artefact development and evaluation within the organisational setting. Consequently, theory informed the artefact design while findings from implementation and evaluation contributed to refinement of the resulting design knowledge.

The research strategy is Design Science Research following the Design Science Research Methodology (DSRM) proposed by Peffers *et al.* (2007). DSR was selected because it supports both artefact creation and generation of transferable design knowledge. Unlike traditional case-study approaches, which primarily seek contextual understanding, DSR enables development and evaluation of purposeful interventions while simultaneously producing reusable design principles.

Finally, a mixed-methods approach was adopted. Quantitative methods were used to analyse pathway intervals, waiting times, utilisation measures, and operational performance metrics, while qualitative methods including stakeholder engagement, requirements elicitation, and artefact evaluation informed design decisions and assessment of utility. The combination of quantitative and qualitative evidence supports both technical evaluation of the artefact and understanding of its operational relevance.

3.1.2 Justification for Design Science Research

Several methodological approaches were considered. A case study methodology would have enabled detailed examination of post-MDT operational processes but would not have provided a structured framework for the design and evaluation of an intervention. Action Research was also considered because of its focus on organisational change and practitioner involvement. However, Action Research typically requires repeated cycles of intervention and measurement within live operational environments. In cancer pathway management, allowing breaches to occur in order to evaluate an intervention would be ethically and operationally undesirable given the known relationship between treatment delays and patient outcomes.

Design Science Research was therefore selected because it provides a rigorous framework for designing, developing, and evaluating an artefact intended to solve a real-world organisational problem while simultaneously generating transferable knowledge. DSR has been widely applied

within healthcare informatics, clinical decision support, and data integration research where new technological artefacts are required to address operational challenges (Harper and Mustafee, 2023; Johnson *et al.*, 2020).

Unlike traditional continuous-improvement approaches such as Plan-Do-Check-Act (PDCA), cancer pathway management cannot rely solely on post hoc measurement of outcomes. A breach of the 31-day or 62-day standard represents a clinically meaningful delay that may adversely affect patient outcomes. Consequently, operational management must emphasise proactive identification and mitigation of risk rather than retrospective evaluation alone. The design objective of the proposed dashboard is therefore not simply to identify breaches after they occur, but to provide leading indicators that enable intervention before delays become clinically significant.

3.2 Design Science Research Framework

The study follows the Design Science Research Methodology (DSRM) proposed by Peffers *et al.* (2007), which structures DSR into six interrelated activities. These activities are applied iteratively rather than strictly linearly.

3.2.1 Problem Identification and Motivation

The problem identification phase was undertaken using a combination of retrospective pathway analysis, operational observations, and literature synthesis. Local cancer waiting time reports and patient tracking data were reviewed to identify recurring delays between MDT outcomes, first oncology consultation, decision-to-treat, and treatment initiation. Findings were triangulated with evidence from the literature review, which demonstrated similar challenges across NHS organisations. This process established the need for an integrated operational analytics solution capable of identifying bottlenecks before breaches occur.

3.2.2 Objectives of a Solution

Solution objectives were derived through a requirements elicitation process combining literature-derived design requirements, operational needs identified through pathway review, and information requirements associated with national cancer waiting time standards. Requirements were grouped into data integration, operational monitoring, decision support, and usability domains.

3.3 Design and Development

The design and development stage focused on constructing an IDR and a suite of operational decision-support dashboards. Development was undertaken iteratively using successive cycles of prototyping, evaluation and refinement with operational stakeholders.

The design and development activity involved two tightly coupled components:

1. Integrated Data Repository (IDR)
2. Decision-Support Dashboard

The IDR adopts a general-model architecture consisting of a staging layer for ETL and harmonisation, a central warehouse, and an application layer for analytics and visualisation, consistent with IDR literature for single-institution deployments (Gagalova *et al.*, 2020).

The dashboard is designed to support operational tasks rather than retrospective reporting, aligning with the role of operational dashboards in healthcare performance management (Buttigieg *et al.*, 2017).

Both components were developed iteratively, guided by explicit design principles derived from prior research and refined during prototyping.

3.3.1 Requirements Elicitation

Requirements were elicited through discussions with operational managers, radiotherapy service managers, and clinical stakeholders involved in post-MDT pathway management.

Initial requirements were gathered through discussions with operational managers within Portsmouth Oncology Centre, review of existing reporting processes, and analysis of pathway-monitoring activities. Particular attention was paid to identifying operational questions that could not be answered efficiently using existing systems.

Key requirements included:

- identification of patients at risk of breaching national waiting-time standards;
- monitoring active oncology referrals and outstanding clinic activity;
- visibility of radiotherapy demand and capacity;
- identification of bottlenecks between referral and treatment initiation; and
- provision of near-real-time operational decision support.

3.3.2 Source System Analysis

Four primary operational datasets were identified, and three were used in the final artefact.

The oncology referral register consisted of a manually maintained SharePoint list used by administrative staff to track incoming referrals and clinical outcomes.

Oncology clinic activity was extracted from the Trust’s Patient Administrative System (PAS), providing appointment, booking and attendance data.

Radiotherapy operational data comprised referral, CT simulation and treatment events extracted from the local Oncology Information System - Aria for Radiation Oncology.

Due to additional projects involving the local Cancer Network, a reliable data source for SACT treatment is not available until January 2027, and was therefore excluded at this time, but design principles allow for including SACT treatments at a later date.

The selected source systems reflected the operational stages of the post-MDT pathway. Together, these systems captured referral management, clinic scheduling, radiotherapy planning, and treatment delivery activities. However, each system was developed independently and used different data structures, identifiers, and reporting mechanisms. Consequently, substantial transformation and harmonisation activities were required before integrated pathway analysis could be performed.

Source	Refresh Frequency	Purpose
Sharepoint Referral Register	Daily	Oncology Referrals
PAS	Weekly	Clinic Activity
Aria RO	Daily	RT workflow

Table 3.1: Source Systems Summary

3.3.3 Data Integration Strategy

A key design decision concerned the method by which operational data would be extracted from heterogeneous source systems. Alternative approaches including direct database connections, JSON-based exports, application programming interfaces (APIs), and manual reporting extracts were considered. However, functionality varied considerably between source systems. While some systems supported direct database access and others allowed JSON exports, comma-separated value (CSV) export functionality was available consistently across all source applications.

A CSV-based ingestion strategy was therefore adopted to maximise interoperability and minimise source-system dependencies. This approach allowed a common extraction process to be implemented across all datasets while avoiding the need for system-specific integration methods.

The resulting architecture prioritised simplicity, portability, and maintainability over real-time integration, which was considered appropriate for an operational dashboard refreshed on a daily basis.

One exception was the Clinic appointment extract from PAS. Due to attempting to capture appointments booked, which acted as a proxy for triage, and also outcome, two reports were provided weekly in the form of Microsoft Excel workbooks containing separate Booking and Appointment worksheets. The revised ETL process automatically ingested both worksheets, preserving booking events and attendance outcomes independently before combining them within the intermediate integration layer. This approach improved representation of the underlying operational workflow while maintaining compatibility with the existing warehouse architecture.

3.3.4 ETL Architecture

Data extraction, transformation and loading processes were developed using Python and PostgreSQL. Individual ETL processes were created for each source dataset and orchestrated through an automated pipeline. The ETL layer was responsible for importing source extracts, performing data quality checks, standardising formats, and loading records into the staging layer of the IDR.

The architecture followed a layered design consisting of:

- Source systems;
- ETL and harmonisation processes;
- Staging tables;
- Intermediate integration views;
- Analytics-ready fact tables; and
- Power BI dashboards.

This separation of concerns simplified troubleshooting, improved transparency, and preserved data lineage throughout the repository.

The architecture deliberately separated raw operational data from analytical representations. Staging tables preserved source extracts in their original form, intermediate views applied business rules and data-quality controls, and fact tables provided analytics-ready datasets for dashboard consumption. This separation reduced coupling between source systems and reporting outputs while supporting auditability and future extension of the repository (See Figure 3.1).

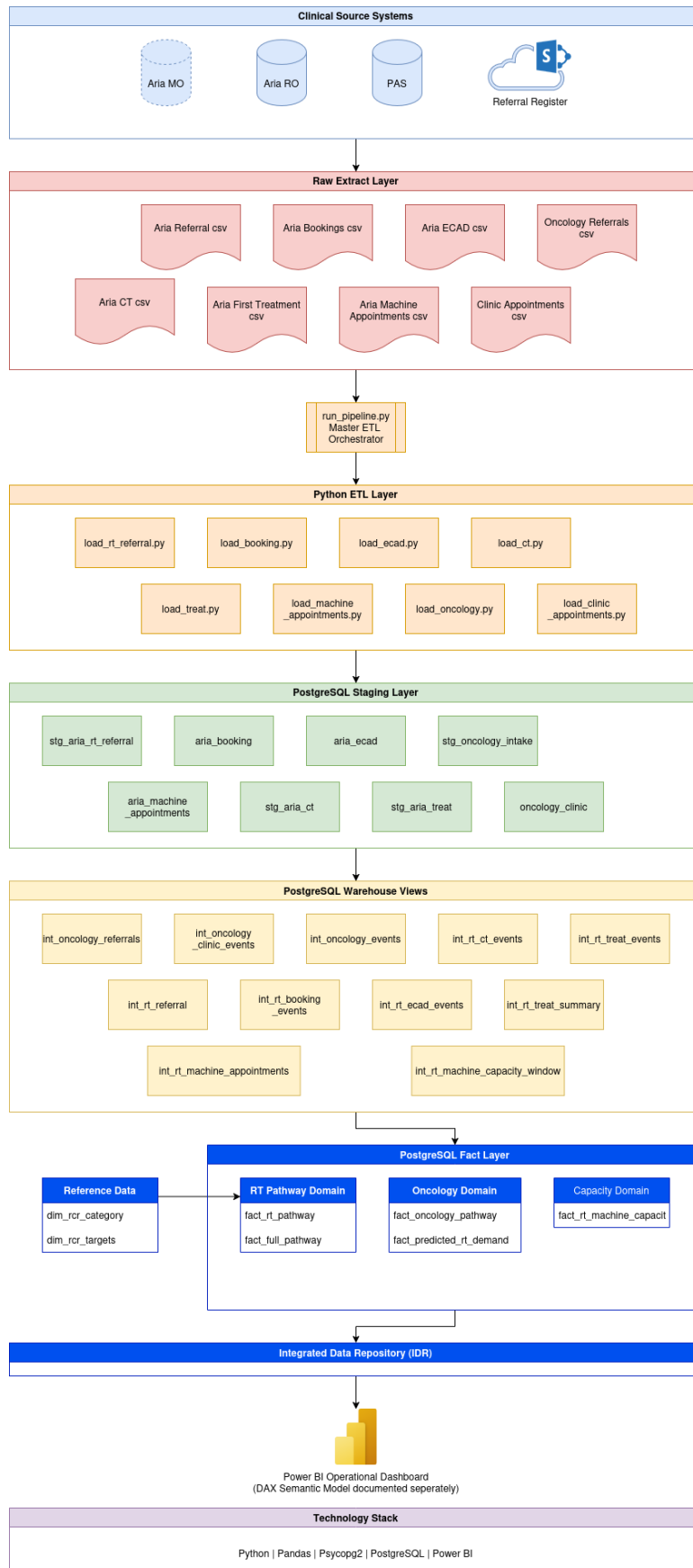


Figure 3.1: Logical Architecture of the IDR

3.3.5 Data Harmonisation and Business Rules

Data harmonisation presented a significant challenge due to variation in identifier quality, inconsistent recording practices, and differences in operational workflows between source systems.

NHS numbers were used as the primary identifier wherever possible and were standardised during transformation to remove formatting inconsistencies. Date formats were similarly harmonised to support accurate interval calculations across systems.

The oncology referral dataset presented particular challenges because it functioned as an operational activity log rather than a controlled referral register. Multiple records were frequently associated with the same oncology episode, reflecting administrative updates, onward referrals, and pathway exceptions. Intermediate transformation views were therefore developed to consolidate records and create a single analytical representation of each referral episode.

Additional business rules were implemented to identify external-provider referrals, duplicate records, deferred follow-up appointments, and non-standard pathways. These rules allowed operational reporting to focus on patients actively progressing through the pathway while preserving the underlying source data for audit purposes.

Following transformation and quality-control activities, analytics-ready fact tables were constructed for oncology pathways, radiotherapy pathways, and integrated referral-to-treatment pathways. These fact tables formed the data source for all dashboard visualisations and performance metrics.

Several previously unrecognised data-quality issues were identified during exploratory analysis. These included duplicate referral records, referrals represented by multiple administrative updates, delayed recording of clinic outcomes, and pathway records relating to patients managed by external providers. The identification and resolution of these issues was necessary to ensure that reported pathway metrics reflected true patient progression rather than administrative artefacts.

3.3.6 Software Engineering and Operational Design

Although the artefact was developed as a research prototype, several software-engineering practices were incorporated to improve reliability and maintainability. Logging was implemented throughout the ETL process to support monitoring, troubleshooting, and auditability. All major extractions, transformation and loading activities produced structured log entries allowing operational issues to be investigated retrospectively.

Database credentials and configuration settings were externalised through environment variables rather than being embedded directly within the application code. This approach aligns

with principles outlined within the Twelve-Factor App methodology, separating configuration from code and reducing the risk of credential exposure.

Together, these practices improved operational robustness and facilitated deployment across development and production environments without modification to application source code.

3.3.7 Pathway Data Model

The purpose of the pathway data model was to transform fragmented operational records into a series of standardised pathway events capable of supporting interval analysis. A key challenge identified during development was individual source systems recorded only portions of the overall oncology pathway. Consequently, no single source provided a complete end-to-end view from referral to treatment initiation.

The pathway model therefore adopted a canonical event-based approach. Individual operational events were mapped to a common set of pathway milestones including referral receipt, oncology triage, first clinic appointment, radiotherapy referral, CT simulation, and treatment commencement. These events were subsequently linked through patient identifiers and temporal matching logic to create an integrated representation of each patient pathway.

From these canonical events, a series of operational intervals were derived, including referral-to-triage, triage-to-clinic, clinic-to-radiotherapy referral, radiotherapy referral-to-CT simulation, and CT simulation-to-treatment. These intervals formed the basis of all dashboard metrics and pathway analyses.

The use of standardised pathway events aligned with Design Principle 3, ensuring that all metrics were derived consistently across source systems and remained traceable to their originating records. Figure 3.2 illustrates the relationship between pathway events, source systems, and derived performance milestones.

3.3.8 Dashboard Development

Three operational dashboards were developed using Microsoft Power BI. Each dashboard was designed to support a distinct operational management task while sharing a common data repository and data model.

The Oncology Referrals dashboard focused on referral management activities and provided visibility of active referrals, clinic waiting times, booking lead times, and unresolved clinic outcomes. The dashboard combined referral-register data with clinic activity to provide a current operational view of patients progressing through the oncology pathway.

The Radiotherapy Service Health dashboard focused on treatment delivery and capacity man-

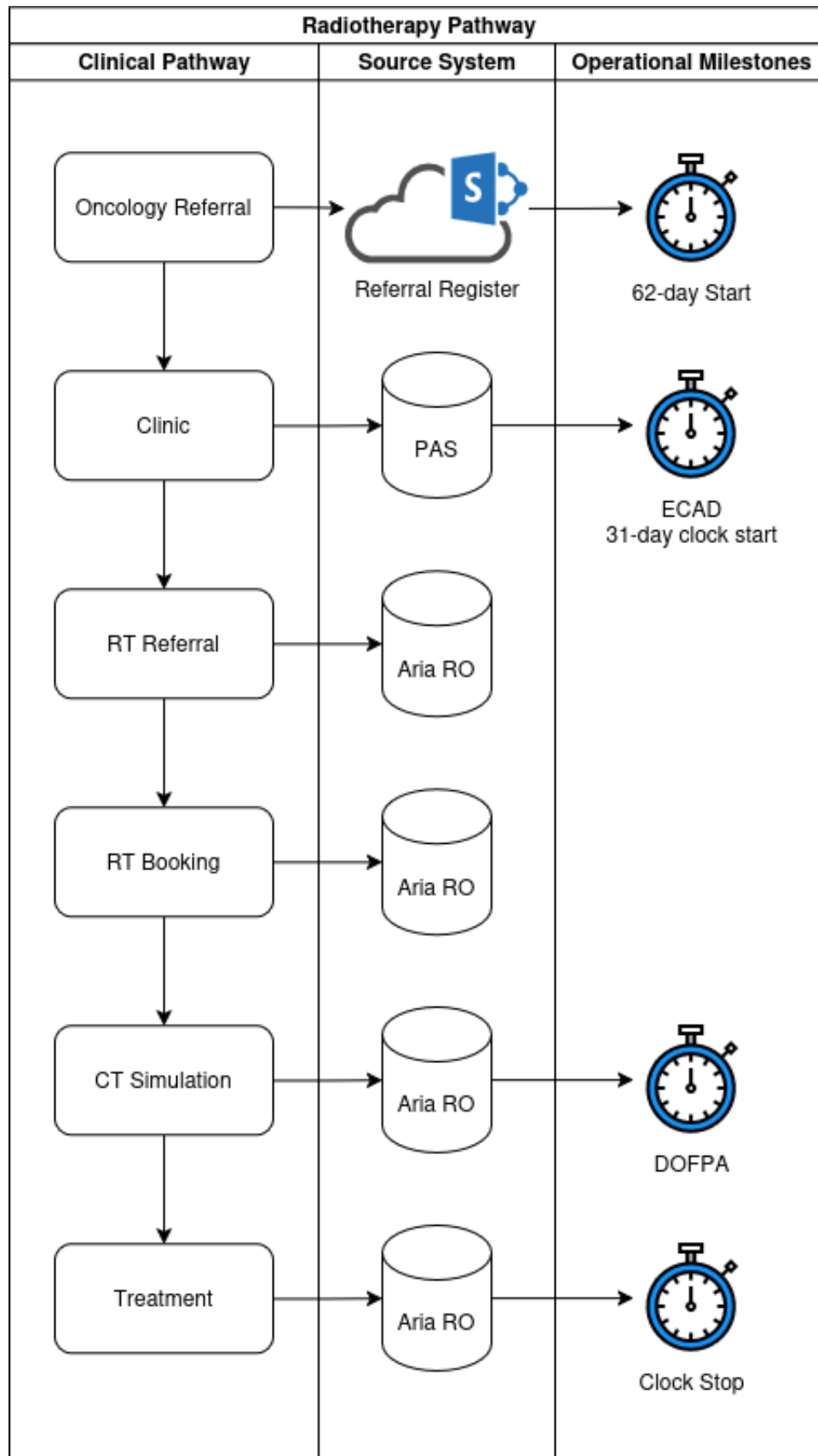


Figure 3.2: Radiotherapy Pathway, Source Systems and Performance Milestones

agement. Operational metrics included active patients, expected future referrals, booking backlog, patient risk categorisation, and projected capacity utilisation. Particular emphasis was placed on presenting risk information alongside capacity and demand metrics to support operational intervention.

The Post-MDT Pathway Overview dashboard integrated data from both oncology and radiotherapy workflows to provide an end-to-end view of patient progression from referral to treatment. The dashboard visualised pathway intervals, operational bottlenecks, and trends in pathway performance over time.

A fourth dashboard addressing Systemic Anti-Cancer Therapy (SACT) pathways was identified during requirements elicitation. However, due to the lack of a reliable and stable SACT data source during the study period, this component was deferred for future development.

Rather than producing a single all-purpose dashboard, functionality was distributed across multiple focused dashboards. This approach aligned with ISO 9241-110 principles by supporting specific operational tasks while reducing visual complexity. Users were therefore presented with information relevant to individual operational decisions rather than a large collection of unrelated metrics.

3.3.9 Iterative Artefact Refinement

Artefact development followed an iterative process consistent with Design Science Research principles. Successive prototypes were demonstrated to operational stakeholders, with feedback being incorporated into later design iterations.

The first development cycle focused on radiotherapy operational management. Initial prototypes concentrated on patient tracking and waiting-time performance; however, stakeholder feedback identified a need for greater visibility of future demand and available treatment capacity. This led to the development of capacity-demand forecasting visuals and booking risk indicators.

The second development cycle focused on oncology referral management. Early versions primarily reported referral volumes and clinic activity. User feedback indicated a stronger operational requirement for active waiting-list management, prompting the addition of patient-group classifications, current booked waits, and referral worklists prioritised by waiting time.

The third development cycle focused on integrated pathway analysis. Stakeholders identified a need to understand where delays accumulated across the patient journey rather than within individual operational areas. This resulted in the development of pathway-stage metrics and a cumulative waterfall visual illustrating time spent between referral, triage, clinic attendance, radiotherapy referral, CT simulation, and treatment initiation.

These iterative refinements ensured that the artefact remained closely aligned with operational requirements while simultaneously providing evidence regarding the effectiveness of the proposed design principles.

3.3.10 Data Quality Management

Exploratory analysis identified several data-quality challenges including duplicate referral records, delayed administrative updates, missing appointment outcomes, and inconsistent coding practices. Rather than modifying source data directly, data-quality controls were implemented within the transformation layer of the repository.

Business rules were used to identify duplicate referral episodes, exclude non-standard pathways from operational waiting-time calculations, and classify clinician-directed surveillance appointments separately from active patient pathways. This approach preserved the fidelity of source data while improving the validity and interpretability of analytical outputs.

Data-quality findings were also treated as a research outcome in their own right, providing insight into the operational limitations of existing pathway-monitoring processes and informing recommendations for future pathway digitisation initiatives.

The discovery of these data-quality issues highlighted limitations in existing operational reporting processes. In particular, the distinction between administrative workflow events and true pathway events emerged as a recurring challenge. This finding informed the design of the pathway data model and represents an additional practical contribution of the study.

Artefact Component	Research Objective
IDR Architecture	Objective 3
Oncology Dashboard	Objective 4
RT Dashboard	Objective 4
Pathway Dashboard	Objective 1
Evaluation Activities	Objective 5
Design Principles	Objective 6

Table 3.2: Artefact to Research Objective mapping

3.4 Design Principles

Design principles represent the design knowledge contribution of this study. They are formulated using a standard DSR structure: *context*, *intervention*, and *rationale*. These initial principles were refined following evaluation.

DP1: Surface Breach Risks as a Leading Indicator

If operational teams must prioritise actions in time-constrained cancer pathways, then dashboards should surface leading indicators of breach risk (e.g. days remaining to breach thresholds), because prospective visibility enables earlier and more effective intervention than retrospective compliance metrics.

DP2: Pair Risk Information with Actionable Capacity Data

If users are expected to act on identified risks, then risk indicators should be presented alongside relevant capacity and demand information, because risk awareness without visible operational levers does not support decision-making.

DP3: Use Canonical, Standard-Aligned Pathway Events

If pathway metrics are to be trusted and comparable, then all intervals should be derived from canonical event definitions aligned to NCWTMDS rules, because inconsistent definitions undermine credibility and evaluation validity.

DP4: Match Data Refresh Cadence to the Action Horizon

If decisions depend on short operational windows, then data refresh frequency should be aligned to the decision horizon (e.g. daily or near-real-time where feasible), because stale information degrades decision quality.

DP5: Preserve Data Provenance and Auditability

If analytics are deployed in regulated healthcare environments, then all derived metrics must maintain transparent provenance to source systems, because trust, governance, and reconciliation are essential for adoption.

DP6: Align Interaction Design with Real Operational Tasks

If dashboards are intended to support complex operational work, then interaction design should reflect users' real task sequences, because task-aligned interfaces reduce cognitive load and improve decision quality (ISO9241, 2020).

These principles directly inform both the IDR architecture and the dashboard design, and serve as evaluation criteria in Section 3.5.

3.5 Evaluation Strategy

Evaluation combines formative and summative methods across four dimensions.

3.5.1 Usability Evaluation

- System Usability Scale (SUS)
- Heuristic review grounded in ISO 9241-110
- Think-aloud walkthroughs

3.5.2 Utility and Effectiveness

- SUS
- Stakeholder feedback
- Design principle validation

3.5.3 Technical Validity

- Reconciliation of dashboard metrics with source SQL queries
- Validation of interval derivations against NCWTMDS rules

3.5.4 Design Knowledge Evaluation

- Synthesis of user feedback and performance outcomes
- Refinement and confirmation of final design principles

3.6 Ethical and Governance Considerations

The study operates within UK GDPR and NHS data governance requirements. All data are de-identified or pseudonymised, with access controls and separation between operational and research use. The project follows university ethics procedures and local Trust information governance guidance.

3.6.1 Research Limitations

- **Single-site study:** The artefact was developed and evaluated within a single NHS Trust, which may limit generalisability.

- **Data quality and availability:** Findings depend on the completeness and accuracy of source systems.
- **Retrospective data:** Retrospective data may not fully represent future operational conditions.
- **Usability sample size:** Evaluation participants are likely to represent a relatively small operational user population.
- **Artificial evaluation environment:** Demonstration scenarios may not fully replicate live decision-making environments.
- **DSR limitation:** Evaluation assesses utility and usability rather than directly measuring improvements in patient outcomes.

3.7 Summary

This chapter has outlined a rigorous, design-science-based methodology that supports both artefact creation and theory-driven knowledge contribution. By embedding explicit design principles within the DSR framework and evaluating the artefact using multiple complementary methods, the study ensures methodological robustness, practical relevance, and academic contribution.

Chapter 4

Results and Analysis

4.1 Introduction

This chapter presents the results obtained from the development and application of the Integrated Data Repository (IDR) and decision-support dashboard suite described in Chapter 3. Following construction of the repository, integrated pathway data were analysed to identify operational bottlenecks, assess service demand and capacity, and evaluate the utility of the artefact for supporting operational decision-making.

Findings are presented according to the research objectives. The chapter begins by summarising repository development and data quality findings before presenting analysis of oncology referrals, radiotherapy operational performance, and end-to-end pathway delays. The chapter concludes by examining the extent to which the developed artefact supports the design principles established in Chapter 3.

4.2 Repository Development Results

The IDR successfully integrated datasets originating from multiple operational systems including oncology referral register, Patient Administration System (PAS), and the Aria radiotherapy management system. Data were extracted, transformed and loaded into a PostgreSQL-based repository using layered architecture consisting of staging, intermediate and analytical reporting layers.

Development of the repository identified several previously unrecognised data quality and interoperability challenges. These included duplicate oncology referral records, inconsistent representation of referral episodes, delayed administrative updates, and differences in event definitions between operational systems.

The use of intermediate integration views allowed business rules to be applied without modifying source data directly. This approach enabled the construction of analytics-ready fact tables capable of supporting operational reporting while preserving auditability and traceability to originating systems.

The repository subsequently became the single source of truth for all dashboard metrics and pathway analysis.

4.3 Oncology Referral Analysis

4.3.1 Referral Workload

Figure 4.1 presents the Oncology Referral Dashboard developed to support management of referral workload, appointment access, and clinic outcome recording.

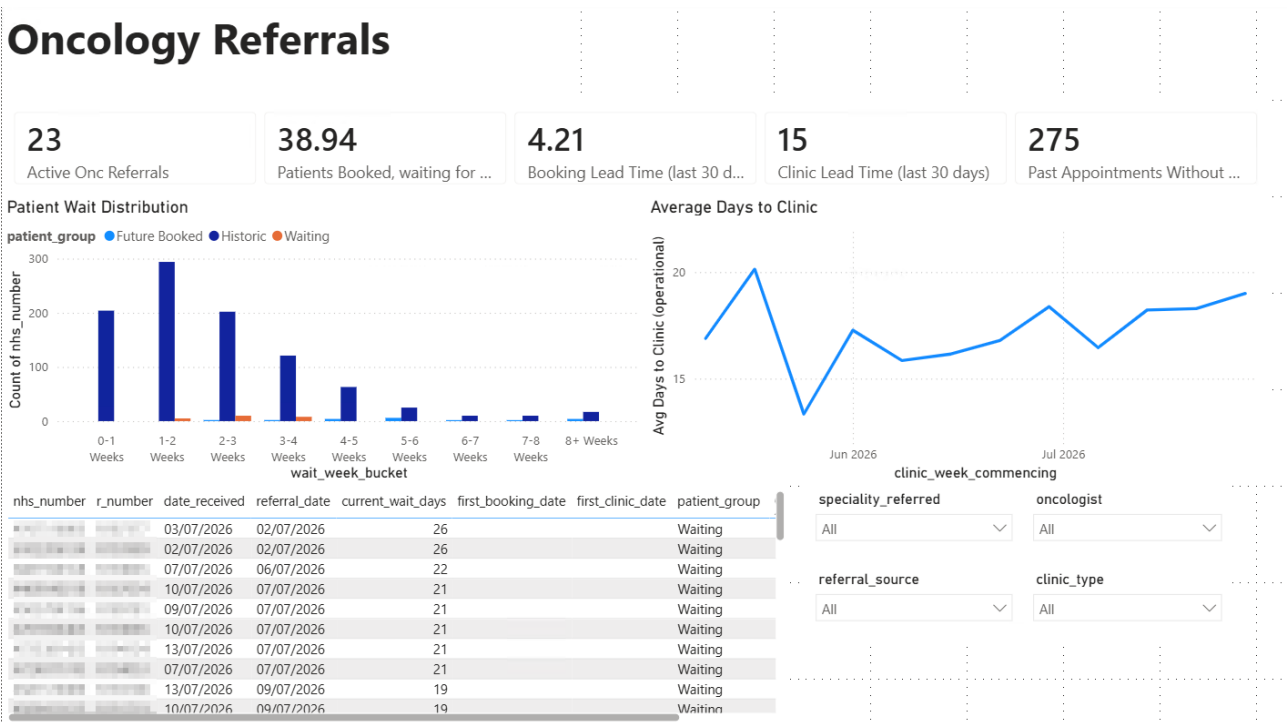


Figure 4.1: Operational Oncology Referral Dashboard

Analysis of the oncology referral dashboard identified 23 active oncology referrals at the time of analysis. While this represents a manageable operational workload, variation was observed in waiting times between referral and clinic attendance.

The distribution of patient waits demonstrated that the majority of patients were clustered within the first three weeks following referral. Smaller numbers of patients remained on the

waiting list beyond six weeks, suggesting that the majority of referrals progressed through the booking process relatively efficiently.

4.3.2 Booking and Clinic Lead Times

The average referral-to-booking interval during the previous 30 days was 4.15 days, indicating that referrals were generally triaged and allocated appointments within one working week.

The median referral-to-clinic interval was 15 days. Trend analysis demonstrated that this value remained relatively stable across the observation period, fluctuating between approximately 13 and 20 days.

These findings suggest that initial referral processing and appointment allocation were not the primary sources of pathway delay. The majority of patients received clinic appointments within a timeframe considerably shorter than national cancer waiting-time thresholds.

4.3.3 Future Booked Waits

Patients holding future clinic appointments had an average booked waiting time of 38.94 days.

This metric should be interpreted carefully, as it does not represent patients awaiting their first oncology appointment following referral. First-appointment access is more accurately reflected by the referral-to-clinic interval, which remained stable at a median of 15 days.

The future booked wait metric incorporates a broader range of appointment types including routine follow-up appointments, treatment reviews, surveillance activity, and result-discussion clinics. Consequently, the longer waiting time observed is likely to reflect the scheduling characteristics of ongoing oncology care rather than delays affecting newly referred patients.

Nevertheless, the measure provides useful visibility of future clinic workload and demonstrates demand already committed within existing clinic schedules. This information may assist operational planning by highlighting the volume of future activity competing for available clinic capacity.

4.3.4 Incomplete Outcome Recording

The dashboard identified 275 historical clinic appointments without recorded outcomes.

This finding represents a significant data-quality issue because pathway progression, attendance analysis, and downstream performance reporting depend upon accurate recording of appointment outcomes. The presence of a large number of unresolved appointments highlights the

importance of integrated operational monitoring and demonstrates how the artefact can reveal process weaknesses that may not be visible through conventional reporting.

4.4 Radiotherapy Service Analysis

4.4.1 Operational Workload

Figure 4.2 illustrates the Radiotherapy Service Health Dashboard developed to support monitoring of utilisation, demand forecasting, and patient risk status.

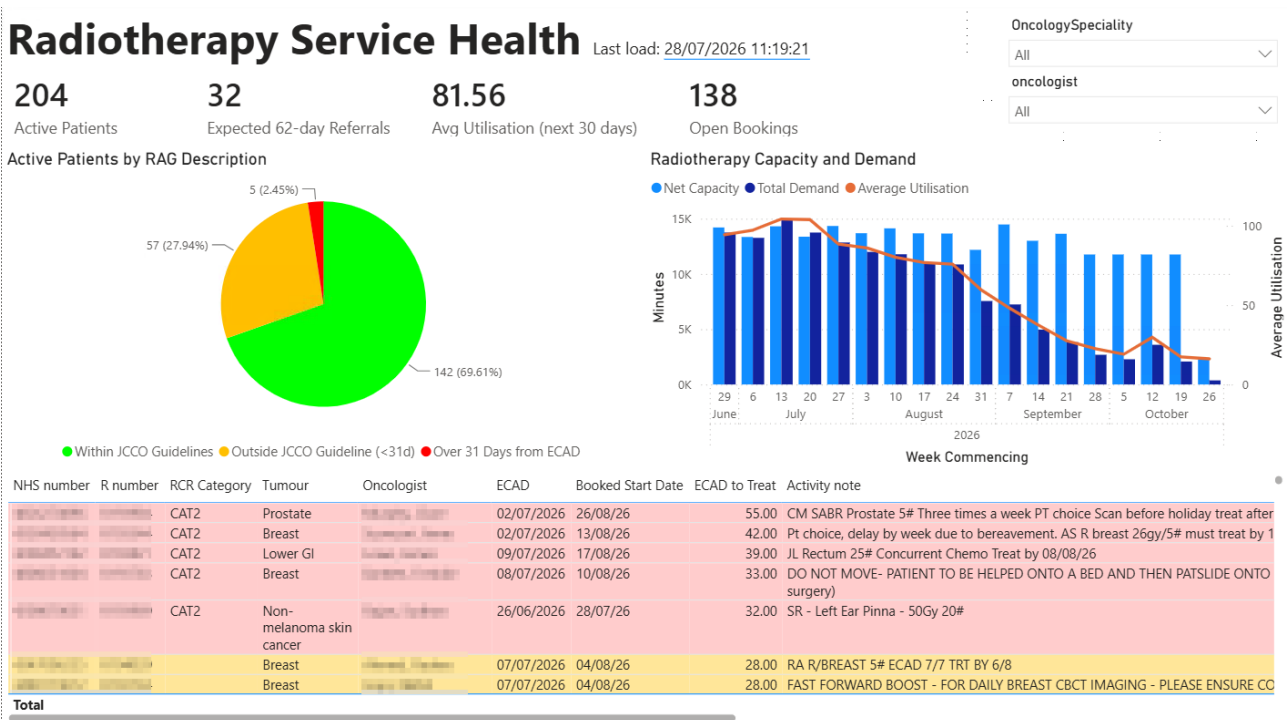


Figure 4.2: Radiotherapy Capacity and Demand Dashboard

The radiotherapy dashboard identified 204 active patients undergoing management within the radiotherapy pathway.

In addition, 32 future referrals were predicted to enter the service under a 62-day pathway, defined as oncology referrals booked into Clinical Oncology clinic. This provides operational managers with advance visibility of future demand and supports proactive planning of treatment capacity.

4.4.2 Capacity Utilisation

Capacity-demand modelling identified an average projected utilisation of 81.56% across the next 30 days, exceeding the departmental planning threshold of 80%. This threshold is used operationally to preserve flexibility for urgent treatments, machine downtime, replanning activity, and other unplanned service pressures.

Forecast utilisation declined across the later periods of the planning horizon. However, this reduction should be interpreted with caution because future demand is inherently less complete than near-term demand. The model includes patients currently booked for treatment and referrals that have already entered the radiotherapy pathway, but does not yet reflect patients who have not progressed sufficiently through upstream stages of the pathway. Consequently, the later forecasting periods are likely to underestimate true future demand.

Despite this limitation, the model provides a useful indication of current service pressure and demonstrates that the radiotherapy service is operating at or slightly above its internal planning threshold.

The significance of this finding should be considered in the context of local service demand. According to National Disease Registration Service (NDRS) benchmarking data, shown in figure 4.3, Portsmouth Hospitals University NHS Trust delivered the highest numbers of radiotherapy attendances per linear accelerator in England for the financial year 2024/25 (NHS Digital, 2026b). Operating close to planned utilisation thresholds therefore reflects both sustained service demand and high equipment productivity. This reinforces the importance of maintaining sufficient operational flexibility to accommodate urgent treatments and unexpected service pressures.

4.4.3 Patient Risk Stratification

Patient-level risk analysis identified:

- 142 patients (69.61%) within JCCO guidelines (Ash *et al.*, 2004).
- 57 patients (27.94%) outside JCCO guidelines, but within NCWT targets (Johnson, 2024).
- 5 patients (2.45%) outside the NCWT 31-day standard.

Although the majority of patients remained within the recommended timeframes, the presence of patients approaching or exceeding target thresholds demonstrates the value of a proactive monitoring approach.

Rather than identifying breaches retrospectively, the dashboard allows operational teams to focus attention on patients most likely to experience future delays.

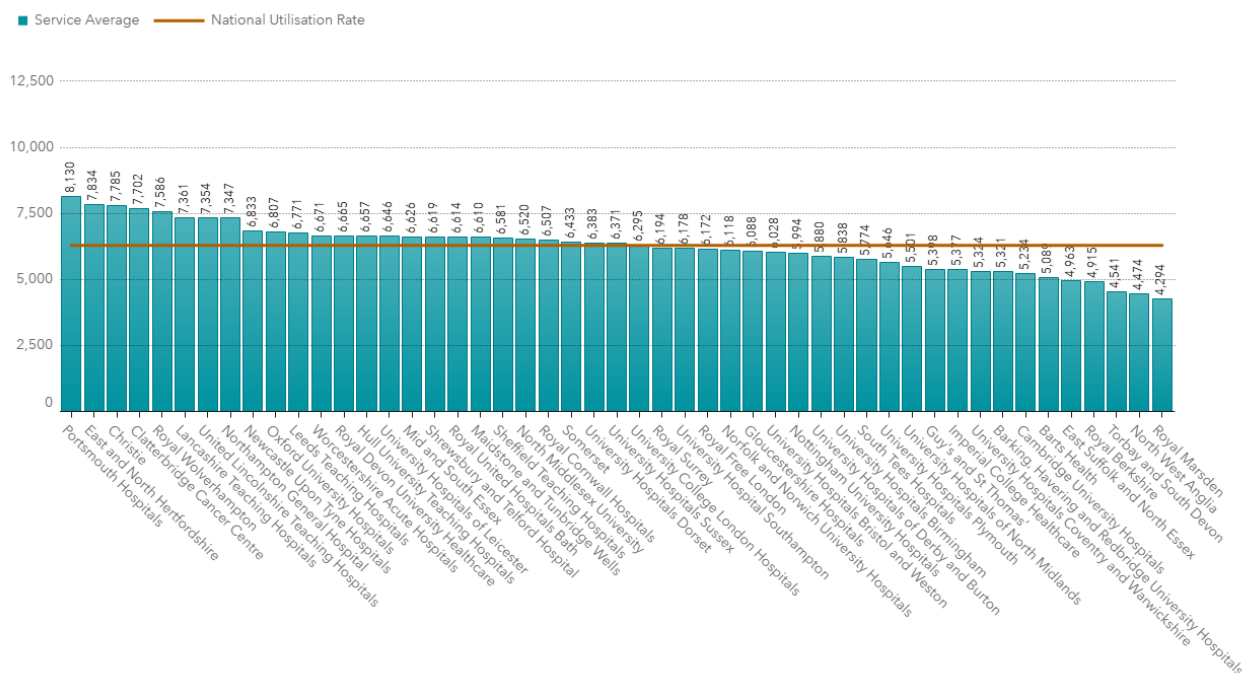


Figure 4.3: Departmental average number of fractions per linac (RAD20) for financial year 24/25 (NHS Digital, 2026b)

4.5 End-to-End Pathway Analysis

4.5.1 Pathway Duration

Figure 4.4 presents the integrated pathway dashboard, combining referral, clinic, and radiotherapy data to visualise pathway progression and identify bottlenecks.

The integrated pathway dashboard (Figure 4.4) enabled analysis of patient progression from referral through to treatment initiation.

The median pathway duration was 37 days and was composed of five principal operational stages:

Stage	Days
Referral →Triage	4
Triage →Clinic	8
Clinic →RT Referral	1
RT Referral →CT	8
CT →Treatment	16
Total	37

Table 4.1: Median duration between canonical events in the Radiotherapy post-MDT pathway

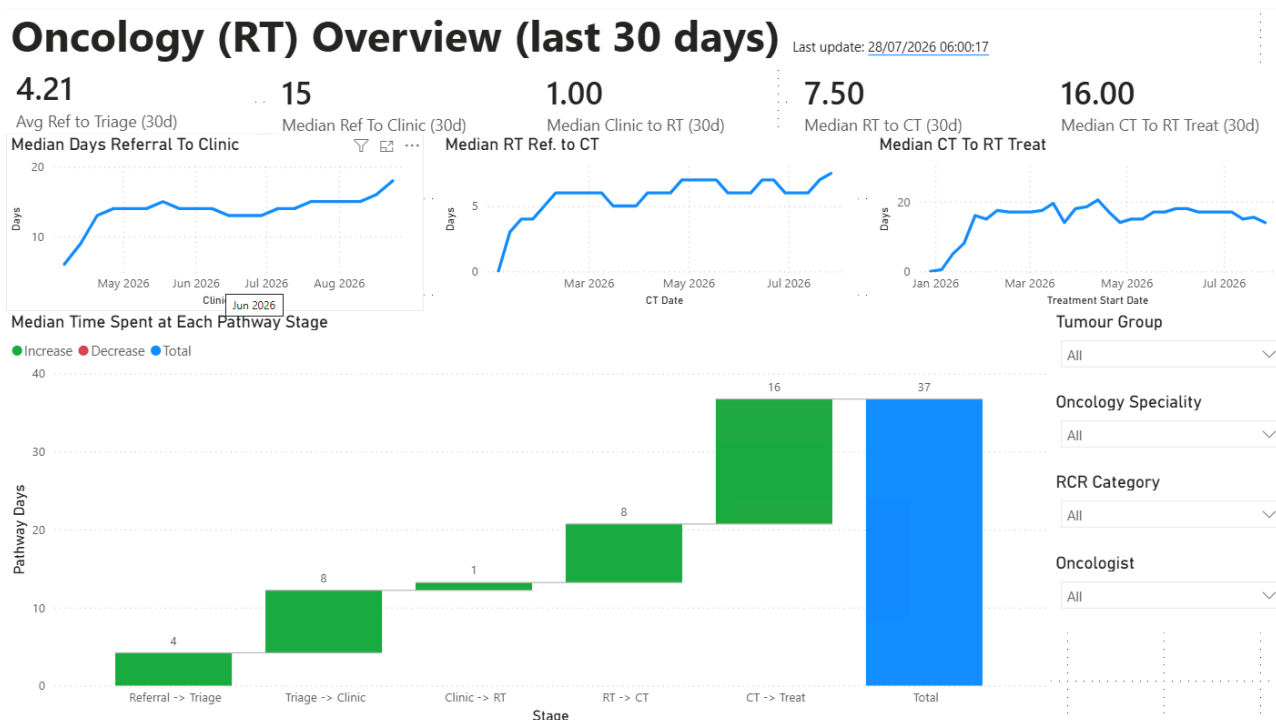


Figure 4.4: Integrated Post-MDT Pathway Dashboard

4.5.2 Bottleneck Identification

The cumulative waterfall visual shown in Figure 4.4 demonstrates the largest delay occurred between CT simulation and treatment commencement, accounting for 16 days of the overall pathway.

The second-largest delay occurred between triage and first clinic attendance, accounting for 8 days.

Together these two stages represented approximately two-thirds of the total pathway duration, suggesting that operational improvement efforts should be concentrated in these areas.

In contrast, the interval between clinic attendance and radiotherapy referral was only one day, indicating that MDT and consultant decision-making processes were not a significant source of delay.

4.5.3 Temporal Trends

Trend analysis demonstrated that:

- Referral-to-clinic performance remained relatively stable.
- RT Referral-to-CT increased from approximately 0 days to 7 days before stabilising.
- CT-to-Treatment remained consistently between 15 and 20 days.

The persistence of the CT-to-treatment interval suggests that a substantial proportion of pathway time is spent between CT simulation and treatment initiation. However, this interval represents an aggregation of several planning and preparation activities rather than a single operational process.

Following CT simulation, patients typically undergo contouring, treatment planning, plan review, quality assurance, and treatment scheduling before treatment can commence. These activities do not currently form part of the canonical pathway events recorded within the repository but nevertheless consume operational time and resources.

Consequently, further investigation is required before this interval can be interpreted solely as a treatment-capacity issue. Opportunities for future work are discussed in Chapter 5.

4.5.4 Research Question 1

The pathway dashboard directly addressed Research Question 1:

Where do delays occur between MDT outcome and treatment?

Analysis demonstrated that delays are concentrated primarily within:

1. CT Simulation → Treatment Start.
2. Triage → Clinic Attendance.

These stages therefore represent the most significant opportunities for operational intervention.

4.6 Operational Evaluation

The dashboard suite was evaluated by three senior operational stakeholders representing different perspectives across the oncology pathway:

- Clinical Director for Oncology
- Radiotherapy Operations Manager
- Lead Treatment Radiographer

Full evaluation results are presented in Appendix D.

Participants were provided with a demonstration of the Oncology referral Dashboard, Radiotherapy Service Health Dashboard, and Post-MDT Pathway Dashboard. Evaluation focussed on usability, perceived operational value, and validation of design principles developed in Chapter 3.

Usability was assessed using the System Usability Scale (SUS). The dashboard suite achieved an average SUS score of 93.3, with individual scores of 95.0, 90.0, and 95.0 respectively. These results indicate a very high perceived usability and suggest that the dashboards were considered intuitive, easy to learn, and straightforward to operate.

All participants agreed that the dashboard improved visibility of post-MDT pathway performance, helped identify patients approaching target breaches, and reduced reliance on manual reporting processes. Two participants strongly agreed the dashboard would be incorporated into routine operational management activities, while the third agreed with the statement.

4.6.1 Perceived Operational Value

Stakeholders reported consistently positive perceptions of the operational value of the artefact. All participants strongly agreed that the dashboards improved visibility of pathway performance and supported identification of patients at risk of breaching performance targets.

The combination of risk monitoring, pathway visibility, and capacity-demand information was considered particularly valuable for supporting operational decision-making. Participants highlighted the potential for the dashboards to reduce reliance on manually generated reports and support more proactive service management.

The Clinical Director identified the pathway waterfall visualisation and filtering by oncologist and speciality as particularly useful, enabling targeted review of pathway bottlenecks. The Radiotherapy Operations Manager highlighted the ability to investigate patients approaching breaches and support planning decisions relating to capacity, overtime, shift extensions, and additional treatment capacity. The Lead Treatment Radiographer identified the detailed patient worklists as the most useful feature, particularly for day-to-day operational management.

4.7 Evaluation of Design Principles

The evaluation suggests that all six design principles were successfully implemented within the final artefact.

Further evaluation was undertaken using structured questions aligned to the six design principles developed in Chapter 3.

Support for DP1 was particularly strong, with all participants strongly agreeing that the dashboard helped identify pathway risks before breaches occurred. Similarly, all participants strongly agreed that capacity information was presented in a useful and actionable manner, providing strong support for DP2.

Design Principle	Evidence from Artefact
DP1: Surface Breach Risks	RAG categorisation and patient risk worklists
DP2: Pair Risk with Capacity	Capacity and demand dashboard visualisations
DP3: Canonical Events	Standardised pathway intervals derived from warehouse fact tables
DP4: Refresh Cadence	Automated ETL pipeline and scheduled refresh processes
DP5: Provenance and Auditability	Staging, intermediate and fact-layer architecture with ETL logging
DP6: Task-Oriented Design	Separate referral, radiotherapy and pathway dashboards aligned to operational workflows

Table 4.2: Artefact evidence of Design Principles

The pathway stages were considered meaningful representations of operational workflow, supporting DP3. Participants generally agreed that refresh frequency was sufficient for operational use, although one participant indicated that more frequent updates could further support treatment-floor operational activities. This finding provides qualified support for DP4.

Participants reported high levels of trust in the dashboard metrics, supporting DP5 and demonstrating the importance of provenance and auditability within healthcare analytics environments.

Finally, all participants agreed that the dashboards aligned well with day-to-day operational activities, supporting DP6 and reinforcing the value of task-oriented dashboard design.

4.8 Summary

The developed IDR successfully integrated multiple operational datasets and enabled detailed analysis of post-MDT oncology workflows. Repository development exposed several data quality challenges while establishing a unified analytical platform for operational decision support.

Analysis of oncology referrals demonstrated relatively efficient referral processing while also highlighting the importance of distinguishing first-appointment activity from ongoing follow-up workload.

End-to-end pathway analysis demonstrated that the greatest delays occur between CT simulation and treatment initiation, with an additional contribution arising between triage and first clinic attendance. These findings provide evidence-based targets for future service improvement initiatives and demonstrate the practical utility of the developed artefact for operational oncology management.

Chapter 5

Discussion and Conclusions

5.1 Introduction

This chapter discusses the findings presented in Chapter 4 in relation to the research objectives, literature review, and design principles developed through the Design Science Research process. The chapter evaluates the extent to which the artefact addressed the identified problem, reflects upon the strengths and limitations of the solution, and considers opportunities for future development.

5.2 Discussion of Research Question 1

Where do delays occur between MDT outcome and treatment?

The primary objective of this study was to improve visibility of delays occurring between MDT outcome and treatment initiation. Analysis of the integrated pathway demonstrated that delays were not distributed evenly across the pathway. Rather, two stages accounted for the majority of elapsed time: triage-to-clinic attendance (8 days) and CT simulation-to-treatment initiation (16 days).

The triage-to-clinic interval represents a period during which referrals are reviewed, prioritised, booked, and ultimately seen by an oncologist. Although the observed interval remained well within national cancer waiting time standards, it nevertheless represented a meaningful proportion of the overall pathway duration. This finding suggests that clinic access remains an important determinant of pathway flow and warrants continued operational monitoring.

A further contribution of the study relates to visibility of oncology referral processing. Prior to development of the repository, no integrated dataset existed that consistently captured referral receipt, triage, and clinic attendance events. Consequently, there was limited organisational

understanding of what constituted a typical or acceptable referral-to-clinic interval. By integrating referral-register and clinic datasets, the repository established an objective baseline against which future operational performance can be measured.

Importantly, the observed median referral-to-clinic interval of 15 days provides the first evidence-based benchmark derived from local operational data. Future iterations of the Oncology Referral Dashboard could build upon this finding by introducing referral-to-clinic risk categories, for example identifying pathways completed within one week as low risk, pathways completed within two weeks as moderate risk, and longer waits as requiring further review. Such thresholds would support proactive monitoring while remaining grounded in observed pathway performance rather than arbitrary targets.

The CT simulation-to-treatment interval represented the largest component of the pathway. However, unlike other pathway stages, this interval does not correspond to a single operational activity. Instead, it encompasses multiple planning processes including target contouring, treatment planning, peer review, quality assurance activities, and treatment booking. Consequently, the current analysis identifies where time accumulates but cannot yet determine which individual planning activities contribute most significantly to delay.

These findings support the argument presented within the literature that aggregated treatment interval metrics provide limited understanding of operational bottlenecks. Studies including Hanna *et al.* (2020), Sharma *et al.* (2016), and Villemure-Poliquin *et al.* (2025) demonstrate the clinical importance of reducing diagnosis-to-treatment intervals. However, such studies typically evaluate the pathway as a single interval and therefore provide limited insight into where delays actually occur.

By decomposing the pathway into canonical operational events, the artefact developed in this study provides a more granular understanding of delay accumulation. This addresses a gap identified in the literature, where operational managers often lack visibility of downstream workflow stages following MDT decision-making (Soukup *et al.*, 2022; Law *et al.*, 2024). Rather than simply measuring compliance with national standards, the repository enables identification of specific pathway stages where operational intervention may have the greatest impact.

5.3 Discussion of Research Question 2

What operational factors are associated with delay?

A secondary objective of this study was to identify operational factors associated with delay. Although the repository was not designed as a predictive modelling platform, several factors emerged during analysis.

First, increasing future clinic waits were observed despite relatively short historic referral-to-

clinic intervals. Patients holding future appointments were waiting substantially longer than patients who had already completed the clinic stage. Although future booked waits were considerably longer than the referral-to-clinic interval, these waits largely reflected planned follow-up and review activity rather than new-patient access. The finding therefore provides greater insight into future clinic workload than referral-processing performance. Further investigation, incorporating clinic templates and appointment-type classifications, would be required before conclusions could be drawn regarding clinic-capacity constraints affecting new referrals.

Second, data-quality issues emerged as an operational factor in their own right. A substantial number of clinic appointments lacked recorded outcomes, while duplicate referral records and inconsistent pathway representations required extensive harmonisation during ETL processing. These findings indicate that administrative recording practices can influence both operational visibility and service-management effectiveness.

The identified data-quality issues were not primarily attributable to technical system failures but were frequently associated with manual administrative processes, retrospective data entry, and the use of locally maintained operational tracking systems. This observation suggests that limitations in operational visibility may arise as much from process design as from analytical capability.

Although implementation of digital referral processes was outside the scope of the present study, the findings provide supporting evidence for ongoing organisational initiatives seeking to replace manual referral-management activities with structured electronic workflows. At the time of writing, the first Urology-to-Oncology electronic referral pathway has recently been introduced within the organisation. Future integration of these structured referral datasets may reduce transcription errors, improve data completeness, and enhance pathway visibility.

More broadly, these findings are aligned with the strategic direction described within the NHS 10 Year Health Plan for England, which advocates a transition from analogue to digital working practices and identifies reduction of administrative burden as a key enabler of service transformation. The data-quality challenges identified during repository development therefore provide a practical example of how structured digital data capture may improve both operational efficiency and analytical capability. (Department of Health and Social Care, 2025)

The importance of these findings is consistent with observations from repository and data-integration literature. Gagalova *et al.* (2020) highlight data harmonisation as a recurring challenge in healthcare repositories, while Varma *et al.* (2026) identify inconsistencies between operational cancer datasets arising from differing definitions and recording practices. Similarly, the present study found that substantial effort was required to reconcile operational data before reliable pathway metrics could be produced.

Third, radiotherapy demand was observed to operate close to departmental planning thresholds. Although utilisation remained within operationally manageable levels, average projected

utilisation exceeded the internal planning threshold of 80%. This finding highlights the importance of balancing efficiency with sufficient flexibility to accommodate urgent treatments and unplanned events.

This finding is consistent with the literature gap identified in Chapter 2, where operational delays are frequently attributed to organisational factors but are rarely examined through integrated operational datasets capable of exposing underlying workflow and information-management challenges.

These findings suggest that operational factors influencing pathway performance extend beyond capacity and scheduling constraints to include data quality, recording practices, and information-management processes.

5.4 Artefact Evaluation

The artefact successfully achieved its primary objective of providing integrated operational visibility across previously disconnected stages of the post-MDT oncology pathway.

From a technical perspective, the repository successfully integrated data from multiple heterogeneous operational systems using a common ETL framework. Automated refresh processes reduced manual data processing requirements while preserving data provenance and auditability.

From an operational perspective, the dashboard suite provided users with visibility of referral activity, clinic performance, radiotherapy capacity, and end-to-end pathway progression. The separation of functionality across multiple dashboards appeared to support task-focused operational workflows without overwhelming users with excessive information.

Stakeholder evaluation further supported this conclusion. The dashboard suite achieved a mean System Usability Scale score of 93.3, indicating very high perceived usability. Participants consistently reported that the dashboards improved visibility of pathway performance, supported proactive identification of patients approaching target breaches, and could reduce reliance on manually compiled operational reports.

Importantly, the development process demonstrated that substantial value can be obtained from relatively lightweight integration architectures. Rather than replacing existing systems, the artefact created a unified analytical layer across existing operational data sources.

The artefact also demonstrated how a relatively lightweight repository can deliver operational value without the complexity associated with large-scale enterprise data warehouses. In contrast to many oncology repositories described within the literature, which are designed primarily for research, genomics, or population-health applications, the present repository was developed

specifically to support operational workflow management. This distinction represents a practical contribution to healthcare informatics and illustrates how established repository principles can be applied within everyday NHS operational environments.

5.5 Evaluation of Design Principles

The Design Principles developed in Chapter 3 were intended to represent transferable knowledge extending beyond the specific implementation used in this study.

DP1 and DP2 received particularly strong support during evaluation. Risk-based patient categorisation and capacity-demand visualisation enabled operational users to identify not only where delays existed but also whether sufficient operational capacity existed to mitigate those risks.

DP3 was supported through the use of canonical pathway events aligned to published cancer waiting time standards. The pathway model enabled consistent interval calculation across multiple source systems and reduced ambiguity associated with locally defined metrics.

DP4 was partially supported. While daily automated refreshes were achieved for the majority of datasets, dependency upon external operational reports limited refresh frequency for some clinic data extracts. Nevertheless, the data remained sufficiently current to support operational decision-making.

DP5 was supported through the separation of staging, transformation, and reporting layers, allowing all reported metrics to be traced back to source records.

DP6 was supported through iterative stakeholder engagement and the creation of task-oriented dashboards aligned with referral management, radiotherapy service management, and pathway performance monitoring respectively.

Collectively, the evaluation suggests that all six design principles remain valid and may have applicability beyond the specific organisational setting studied.

The design principles developed throughout this study are consistent with observations from dashboard, usability, and operational analytics literature. In particular, the importance of proactive risk identification aligns with concerns raised in the literature regarding the limitations of retrospective performance reporting. Similarly, the emphasis on capacity-aware decision support reflects the need for actionable operational intelligence rather than passive presentation of performance data alone.

The evaluation therefore provides preliminary evidence that these principles may have applicability beyond the specific organisational context investigated during this study.

5.6 Reflections and Lessons Learned

Several important lessons emerged during the development process.

The most significant lesson related to data quality. Initial assumptions suggested that the primary challenge would involve technical integration of multiple systems. In practice, the greater challenge involved understanding operational processes and interpreting inconsistently recorded data.

The project also highlighted the value of Design Science Research for healthcare informatics projects. The iterative nature of the methodology enabled technical development to remain aligned with operational requirements while simultaneously generating transferable design knowledge.

An additional lesson concerned the relationship between operational processes and data quality. Initially, data-quality issues were viewed primarily as technical challenges requiring harmonisation within the repository. However, analysis demonstrated that many issues originated from manual administrative workflows rather than technical limitations. This shifted the focus from data correction towards consideration of process redesign and digital data capture, highlighting the close relationship between operational practice and analytical quality.

Finally, the study reinforced the importance of stakeholder engagement. Several dashboard features and pathway metrics evolved considerably following operational feedback. This iterative refinement improved both the utility and relevance of the final artefact.

5.7 Limitations

Several limitations should be considered when interpreting the findings.

First, the study was conducted within a single NHS organisation and therefore reflects local processes, systems, and operational practices.

Second, the repository relies upon the quality and completeness of source-system data. Although significant harmonisation and validation activities were undertaken, residual inaccuracies within source datasets may still influence results.

Third, the CT simulation-to-treatment interval represents an aggregation of multiple planning and operational activities. While the repository successfully identifies this stage as a significant contributor to overall pathway duration, the current data model cannot determine precisely how time is distributed between individual planning processes.

Finally, evaluation focused primarily on operational utility and usability rather than direct

patient outcomes. Consequently, although the artefact provides improved operational visibility, the extent to which it contributes to improved clinical outcomes remains outside the scope of the present study.

5.8 Future Work

Stakeholder feedback was broadly consistent with the future development opportunities identified through pathway analysis. Participants highlighted the importance of integrating SACT datasets, improving clinic-capacity visibility, and further decomposing the CT simulation-to-treatment interval. Additional suggestions included visibility of emergency inpatient referrals and greater understanding of the factors contributing to reduced machine utilisation. These findings provide further support for the proposed future-work programme.

Planned future work, in order of project maturity:

5.8.1 Integration of Digital Referral Pathways

Several data-quality issues identified during repository development were associated with manual data entry, transcription, and the use of locally maintained operational tracking systems. Structured digital referral processes have the potential to improve data completeness, reduce transcription errors, and increase consistency of pathway-event recording.

At the time of writing, organisational work is already underway to replace manual referral processes with electronic referral forms, with the first Urology-to-Oncology referral pathway becoming operational during the final stages of this project. This development represents a positive step towards capturing structured referral data at source and reducing reliance on retrospective administrative data entry.

Future work should focus on integrating these emerging digital referral datasets into the Integrated Data Repository. Similar to the planned integration of Systemic Anti-Cancer Therapy (SACT) data following the regional system upgrade, incorporation of structured referral data would improve data quality, enhance pathway visibility, and reduce the need for downstream data harmonisation. As additional tumour pathways adopt electronic referral processes, the repository architecture developed during this study provides a foundation for incorporating these new data sources into a unified operational analytics platform.

The recommendation to integrate emerging digital referral datasets is also consistent with the wider strategic direction described within the NHS 10 Year Health Plan for England (Department of Health and Social Care, 2025). In particular, the plan identifies a transition from analogue to digital ways of working and the reduction of administrative burden through im-

proved use of technology. The data-quality issues identified during this study provide practical evidence of the challenges associated with manually maintained operational processes and further support the case for structured digital data capture at source.

Stakeholder feedback identified an additional benefit of integrating structured referral datasets. Because the repository now provides a quantitative baseline of referral-processing performance, future evaluations will be able to compare pathway behaviour before and after implementation of electronic referral pathways. This creates an opportunity to measure whether structured referral capture improves data quality, reduces administrative processing time, and shortens referral-to-clinic intervals. Consequently, the repository may provide a useful evaluation framework for assessing the impact of ongoing digital-transformation initiatives.

5.8.2 CT-to-Treatment Decomposition

The current pathway model treats CT simulation-to-treatment as a single interval. Future repository development should introduce additional canonical events representing contouring, treatment planning, peer review, quality assurance, and treatment booking activities. This would provide a more granular understanding of where delays occur within the treatment-planning process.

5.8.3 Clinic Capacity Modelling

Future development should incorporate clinic-capacity modelling and appointment-type classification to better understand demand distribution across new-patient, follow-up, and review clinics. This would provide greater visibility of capacity utilisation and allow more detailed investigation of factors influencing access to first oncology appointments.

5.8.4 SACT Integration

A key future enhancement involves integration of Systemic Anti-Cancer Therapy (SACT) data following completion of the planned Cancer Network-wide system upgrade. Incorporating SACT treatment events would enable end-to-end analysis of all major treatment modalities and provide a more comprehensive representation of post-MDT oncology pathways.

5.8.5 Predictive Analytics

The repository architecture also provides a foundation for future predictive analytics. Potential applications include forecasting breach risk, modelling future demand, and identifying patients

likely to experience pathway delays before breaches occur.

5.9 Conclusion

This study has demonstrated that meaningful operational insight can be generated through the integration of existing oncology datasets within a lightweight Integrated Data Repository. Using a Design Science Research methodology, the project successfully developed and evaluated an artefact capable of improving visibility of post-MDT pathway performance.

The repository identified that delays are concentrated primarily within the triage-to-clinic and CT simulation-to-treatment stages of the pathway. Furthermore, the project demonstrated that significant operational value can be achieved through relatively modest data integration and analytics capabilities without requiring replacement of existing clinical systems.

Beyond the artefact itself, the study contributes a set of evaluated design principles for operational oncology analytics that may be transferable to other healthcare organisations facing similar challenges relating to fragmented data and pathway visibility.

In contrast to many oncology data repositories described within the literature, the artefact was designed specifically to support operational workflow management rather than research data reuse. The study therefore contributes both a practical solution and evidence regarding the application of Design Science Research within NHS operational analytics.

The findings therefore contribute both practical value for operational cancer services and academic value through the generation of reusable design knowledge for healthcare operational analytics.

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Appendix A

Integrated Data Repository Architecture

A.1 Repository Layers

The Integrated Data Repository (IDR) adopted a layered architecture consisting of source systems, staging tables, intermediate integration views, analytical fact tables, and reporting dashboards. This design separated raw operational data from analytical representations while maintaining data provenance and traceability.

A.2 Primary Repository Components

Component	Purpose
Staging Layer	Preservation of raw source extracts and audit trail.
Integration Layer	Application of business rules and data harmonisation processes.
Fact Tables	Analytics-ready datasets used for operational reporting.
Power BI Layer	Operational dashboards and data visualisation.

Table A.1: Repository Architecture Components

Source	Data Type	Purpose
Oncology Referral Register	SharePoint List	Oncology referral management
Patient Administration System (PAS)	Operational Reports	Clinic bookings and attendance
Aria OIS	CSV Extracts	Radiotherapy referral, CT and treatment activity

Table A.2: Primary Data Sources

A.3 Source Systems

A.4 Logical Architecture

Figure 3.1 in Chapter 3 illustrates the logical architecture of the repository.

Appendix B

Data Dictionary

Column	Description
oncology_referral_date	Date oncology referral received.
oncology_booking_date	Date first oncology appointment booked
oncology_clinic_date	Date first oncology clinic attended.
rt_referral_date	Date referral received by radiotherapy service.
ecad_referral_date	Earliest clinically agreed decision-to-treat date.
ct_date	CT simulation date.
first_completed_treat_date	First completed radiotherapy fraction.
days_onc_to_clinic	Days from referral to first oncology clinic.
days_rt_to_ct	Days from radiotherapy referral to CT simulation.
days_ct_to_treat	Days from CT simulation to treatment start.
operational_risk_group	Current operational risk classification.
booking_rag_status	Booking risk category.

Table B.1: Data Dictionary

Appendix C

Example ETL and Business Rules

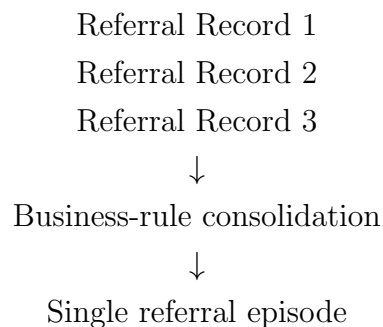
C.1 Identifier Standardisation

NHS numbers were standardised during transformation to remove formatting inconsistencies and improve matching across source systems.

```
REPLACE(nhs_number,' ','')
```

C.2 Referral Episode Consolidation

The oncology referral register contained multiple administrative records relating to a single referral episode. Business rules were therefore applied to consolidate multiple records into a single analytical representation.



C.3 Incomplete Clinic Outcome Identification

One data-quality issue identified during repository development concerned clinic appointments lacking recorded outcomes. These records were flagged during transformation and subsequently reported through operational dashboards.

```
CASE
  WHEN appointment_outcome IS NULL
  THEN 'Missing Outcome'
END
```

C.4 Radiotherapy Pathway Status Classification

```
CASE
  WHEN has_completed = 1 THEN 'Treated'
  WHEN has_cancelled = 1 THEN 'Closed - Cancelled'
  WHEN has_active_booking = 1 THEN 'Active'
  WHEN booking_completed_date IS NOT NULL
    THEN 'Closed - No Treatment'
  ELSE 'Awaiting Booking'
END
```

C.5 Canonical Pathway Events

The repository transformed operational events into standardised pathway milestones:

1. Referral Received
2. Oncology Triage
3. First Oncology Clinic
4. Radiotherapy Referral
5. CT Simulation
6. Treatment Commencement

These events provided the basis for interval calculations and pathway analysis.

Appendix D

Dashboard Evaluation Results

D.1 Participants

Three operational stakeholders participated in the evaluation of the dashboard suite.

Participant	Role
P1	Clinical Oncology Director
P2	Radiotherapy Operations Manager
P3	Lead Treatment Radiographer

Table D.1: Evaluation Participants

D.2 System Usability Scale Results

Participant	SUS Score
P1	95.0
P2	90.0
P3	95.0
Mean	93.3

Table D.2: System Usability Scale (SUS) Results

The dashboard suite achieved a mean SUS score of 93.3, indicating a very high level of perceived usability.

D.3 Operational Value Assessment

Participants rated the following statements using a five-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree).

Statement	Mean Score
The dashboard improves visibility of post-MDT pathway performance	5.0
The dashboard helps identify patients at risk of target breaches	5.0
Displaying capacity and demand information supports operational decision-making	4.7
The dashboard reduces reliance on manual reporting processes	5.0
I would incorporate this dashboard into routine operational management activities	4.7

Table D.3: Operational Value Assessment

D.4 Design Principle Validation

Statement	Mean Score
The dashboard helps identify risks before breaches occur	5.0
Capacity information is presented in a useful and actionable way	5.0
The pathway stages reflect meaningful operational events	4.7
Information is refreshed frequently enough for operational use	4.3
I trust the dashboard metrics	4.0
The dashboard supports my day-to-day operational tasks	4.3

Table D.4: Design Principle Evaluation Results

D.5 Qualitative Feedback

Theme	Feedback
Most Useful Features	Waterfall visualisation of pathway delays; filtering by oncologist and speciality; patient worklists and breach monitoring.
Missing Information	SACT data; clinic utilisation; visibility of emergency in-patient referrals to radiotherapy; utilisation breakdown by machine availability versus service demand.
Operational Decisions Supported	Referral monitoring; patient tracking; bottleneck identification; overtime planning; shift extensions; additional linear accelerator capacity; capacity planning based on expected referrals.
Future Enhancements	Clinic-capacity modelling; CT-to-treatment decomposition; utilisation analysis; integration of structured electronic referrals and SACT datasets.
General Comments	“Very good. Can’t wait to start using it.” “Looks good.” Electronic referral implementation was identified as a useful future data source for before-and-after evaluation studies.

Table D.5: Summary of Qualitative Feedback